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SURVEY

Green Cybersecurity: Leveraging AI, ML, and LLMs to Optimize Energy, Threat Detection, and Sustainability Frameworks

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ABSTRACT The rapid advancement of digital technologies has significantly increased energy consumption and carbon emissions, raising critical concerns about environmental sustainability. *Green cybersecurity*, which integrates environmental sustainability goals into security processes, is becoming increasingly vital as AI systems contribute to growing energy use and carbon dioxide emissions. This paper introduces the concept of green cybersecurity by examining how the integration of Artificial Intelligence (AI), Machine Learning (ML), and Large Language Models (LLMs) can simultaneously enhance cybersecurity practices while reducing their environmental impact. Through a systematic review of peer-reviewed literature, technical reports, and case studies from 2019 to 2025, we explore the potential of AI and ML to optimize energy usage, minimize electronic waste, and support the development of environmentally sustainable cybersecurity frameworks. Our findings indicate that AI and ML can drive advancements in resource management, predictive threat detection, automated response, and real-time monitoring. To assess these opportunities, we draw upon multiple theoretical frameworks, including the Systems Theory Model, Cybersecurity Maturity Model (CMM), and the UN Sustainable Development Goals (SDG) Framework, which collectively emphasize the socio-technical and ethical dimensions of AI-powered cybersecurity. We also address key challenges such as high computational demands, integration complexity, lifecycle environmental impacts, and ethical concerns related to data privacy and algorithmic transparency. This study underscores the need for interdisciplinary collaboration to develop security solutions that are both technologically robust and environmentally responsible.

INDEX TERMS Artificial intelligence, machine learning, large language models, cybersecurity maturity model, energy optimization, sustainable development goals.

I. INTRODUCTION

Environmental sustainability has emerged as a critical issue in an era where digital technology is transforming industries and societies. In the tech industries, the rapid growth of technology in recent years has brought about significant benefits across various industries, including improved efficiency, increased connectivity, and remarkable access to information. As demand for this technology increases, the need to protect information and data from cyber threats and attacks arises, leading to the adoption of cybersecurity [1]. Cybersecurity has become an essential tool to ensure the

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safety and integrity of information [2]. However, the growing number of cybersecurity measures, especially those that require high energy for processing, such as AI-driven threat detection and large-scale data computation, have resulted in high energy consumption, electronic waste, and carbon emissions [3]. In fact, the growing challenge of green cybersecurity is underscored by the hidden environmental costs of our digital infrastructure. In today's digital age, cybersecurity is essential for protecting sensitive information, maintaining privacy, and ensuring the integrity of systems and networks. As individuals, businesses, and governments increasingly rely on digital technologies, the risk of cyber threats such as data breaches, ransomware attacks, and identity theft continues to grow. Effective cybersecurity measures help prevent unauthorized access, detect and respond to threats, and safeguard critical infrastructure. Without robust cybersecurity, the consequences can include financial loss, reputational damage, legal liabilities, and disruption of essential services. There exists a large system of data centers that require significant amounts of energy, materials, and power to operate.

According to the International Energy Agency (IEA), data centers now use about 1% to 2% of the world's electricity. Although this may seem modest, large data centers can consume more than 100 MW of power, enough to energize between 350,000 and 400,000 electric cars annually [4]. In some regions, the figures are even more striking; for example, data centers in Ireland account for more than 20% of the country's total electricity, while those in the United States, China and the European Union consume between 2% and 4% of the total electricity [4]. The environmental burden is compounded by the energy-intensive nature of artificial intelligence (AI). Training advanced models such as GPT-3 can emit approximately 626,000 pounds (284 metric tons) of carbon dioxide, comparable to the lifetime emissions of five average passenger vehicles [5]. Likewise, training a single BERT base model can use energy equivalent to that consumed on a transcontinental flight from New York to San Francisco, and the demand for computing power in AI has increased by approximately 300,000 times in just six years [6]. Furthermore, data centers in the United States used about 626 billion liters of water in 2020, an amount comparable to the annual water consumption of more than 2 million households, raising concerns in regions already facing water scarcity [7].

In order to address these critical challenges, the concept of green cybersecurity is emerging as a promising solution. Green cybersecurity involves the integration of environmental sustainability goals into cybersecurity practice with a focus on ensuring the reduction of energy consumption, minimizing carbon footprints, and promoting environmentally friendly technologies [8]. The integration of Artificial Intelligence (AI) and Machine Learning (ML) into cybersecurity practices provides strong tools to enhance cybersecurity operations and energy conservation. AI and ML have been instrumental in transforming the field of cybersecurity; these technologies enable the development of advanced algorithms that can detect and mitigate cyber threats [9]. AI and ML can analyze vast amounts of data in a short time, identify patterns and anomalies, and can also automate responses to potential threats [10]. This approach not only ensures that cybersecurity measures are robust and effective, but also contributes to global sustainability efforts and potentially lowers operational costs through improved energy efficiency. By adopting these technologies, organizations can limit carbon emissions and manage electronic waste (e-waste) associated with cybersecurity operations. Research indicates that implementing energy-efficient cybersecurity measures not only promotes resource conservation but also improves security [11].

In addition, many companies are increasingly making sustainability a central part of their business activities. For example, Google reached a key achievement by fully powering its global operations with renewable energy in 2021 [12]. Microsoft has also pledged to remove more carbon from the environment than it produces by 2030, with plans to transition its data centers to renewable energy by 2025 [13]. Similarly, major data center providers like Amazon Web Services (AWS), Microsoft, Google, and Facebook have committed to replenishing more water than they use by 2030. This includes prioritizing efficient water management and responsible water supply [7]. These efforts highlight a clear movement toward a more environmentally responsible digital future. Moreover, the integration of green principles into cybersecurity frameworks can also lead to innovative solutions that address both security and environmental challenges, promoting a more sustainable digital future [14].

In this review paper, we will explore the intersection of AI, ML, and cybersecurity within the context of environmental sustainability. It also examines the current state of green cybersecurity and highlights how AI and machine learning can be used to develop more green cybersecurity solutions.

A. REVIEW OBJECTIVES

AI and ML have also gained popularity in promoting environmental sustainability in various domains, from healthcare and finance to transportation and education. However, despite the wealth of excellent surveys available on AI and cybersecurity, there remains a notable gap in the literature. Specifically, there is a lack of comprehensive reviews that focus on the integration of AI and ML within a wide variety of green cybersecurity applications. This paper aims to provide a comprehensive overview of the current state of green cybersecurity, examine the environmental implications of cybersecurity operations, and identify the challenges and opportunities associated with the integration of AI, ML, and LLM into green cybersecurity strategies. The demand for AI and ML-driven services is projected to increase 30 to 40% annually over the next 5 to 10 years [15]. Even though there is growing concern for sustainability

and AI-based cybersecurity, there is no available literature with a harmonized view for exploring their environmental convergence. The paper bridges that by combining technical practices (e.g., hardware acceleration, model optimization) and frameworks (e.g., CMM, SDGs) with a holistic strategy to achieve environmentally sustainable cybersecurity systems. It is crucial to ask:

- What are the dangers of the tremendous growth of AI and Ml in our environment?
- What can be done to mitigate these dangers and meet our climate and sustainability goals in the digital age?

The specific objectives of this survey are the following.

- Review the environmental impact of cybersecurity operations.
- Identify key areas where AI and machine learning can simultaneously improve cybersecurity operations and promote environmental sustainability.
- Identify challenges and research gaps in the application of AI and ML in cybersecurity.
- Identify future research directions and practical implementations for integrating AI and ML into green cybersecurity practices.

B. PRIOR WORK AND SHORTCOMINGS

The use of AI and ML for sustainability and cybersecurity has become one of the most discussed issues in the last few years. A systematic overview of current research demonstrates how AI is used for environmental sustainability before progressing to how it is utilized in cybersecurity, ultimately bridging the two via the concept of Green Cybersecurity.

The applications of AI for environmental sustainability are extensive and varied and include ecosystem monitoring, natural resource management, and climate change mitigation [16]. Research work in [17] provides a comprehensive report of AI applications in key areas of sustainability, including water management, energy efficiency, conservation of biodiversity, and transportation. The authors in [18] study the potential of AI to enable sustainable resource management while mitigating its environmental footprint.

Green computing has become an important component of sustainability. A study by Harmon and Auseklis has investigated the areas of energy-efficient hardware, energy-efficient cooling systems, and electronic waste disposal in a sustainable way [19], [20]. These questions emphasize that the implementation of sustainable IT practices, such as efficient power management and recycling processes, minimizes carbon footprints and improves system resilience. Likewise, Guha Paul et al. emphasize the advantages of sustainable IT solutions to improve the overall efficiency of computer systems [21]. With growing dependence on digital infrastructure for sustainability, cybersecurity's role in the environmental protection sector has become even more significant. The work of Sulich et al. and Ntuli et al. points out that the Environmental Goods and Services sector, among others, requires strong cybersecurity defenses to protect processes that are vital to effective environmental management [22], [23]. It is this phenomenon that has given rise to the concept of Green Cybersecurity, where robust digital structures serve as catalysts for sustainability initiatives. As noted by Stanescu and Ewim et al., the adoption of cybersecurity in the form of sustainable structures guarantees the shift from a linear economy to a circular economy [24].

Obasi et al. and Sargsyan reinforce this perspective, demonstrating the way that cybersecurity aids in progressing toward enhancing sustainability through the protection of the technologies used for nature conservation [14], [25]. Currently, the direction of research has increasingly moved towards the intersection of AI, ML, and cybersecurity. Vallabhaneni [26] highlights the way AI-driven cybersecurity is not just protective against cyber threats, but also contributes to sustainable digital growth. Durlik et al.'s [27] work also explains in detail the challenge of AI adoption in sustainable industries, particularly shipping and ports, and offers solutions in terms of edge computing and IoT to address the problems of data security and connectivity.

Furthermore, Roshanaei et al. [28] discuss the dual role of AI in cybersecurity, quantifying both its potential to strengthen security defenses and vulnerabilities. Their work uses historical trends, technology assessments, and predictive modeling to chart the evolving impact of AI on cybersecurity.

Extending the security-sustainability nexus, Uzondu and Lele's research highlights the necessity of end-to-end cybersecurity systems in smart environmental applications to mitigate looming cyber attacks [29]. Similarly, Dede et al. and Obasi et al. highlight the role of cybersecurity in supporting the United Nations Sustainable Development Goals (SDGs) by providing assurance for the resilience and integrity of critical systems [14], [30].

The intersection of artificial intelligence, machine learning, sustainability, and cybersecurity presents a host of opportunities and threats. A strategy that integrates technological advancement with robust security measures and moral principles is essential. Through the development of secure and sustainable digital infrastructures, this approach promotes cooperation among different sectors and the responsible adoption of artificial intelligence in cybersecurity, while at the same time defending environmental and financial interests. A clearer conceptual separation of green IT, AI/ML-enabled cybersecurity, and their intersection is necessary to understand green cybersecurity. Green IT revolves around minimizing environmental influence through energy-efficient infrastructure, virtualization, and sustainable hardware strategies. On the other hand, AI/ML-enabled cybersecurity is about anomaly detection, real-time threat prediction, and self-ruling response systems. However, their combination in green cybersecurity seeks to achieve robust cyber defense with a minimum environmental footprint. Model pruning, edge computing, and accelerator-based architectures are some of such instances. Their combination requires lifecycle analyzes with considerations of security effectiveness and environmental cost considerations in tandem.

C. CONTRIBUTIONS OF THIS SURVEY

Comprehensive Integration of AI, ML, and Green Cybersecurity: Most previous studies focus on individual aspects such as AI for sustainability [17], [18], green computing practices [19], [20], or AI-driven cybersecurity [26], [28] separately. Unlike these separate approaches, our paper shows how AI, ML, and LLM can work together to improve cybersecurity practices while reducing environmental harm through a single framework.

1) **Addressing Environmental Impacts of Cybersecurity Operations:** While prior works cover green IT practices and AI applications in sustainability separately, no complete analysis exists that evaluates the environmental costs of cybersecurity operations themselves. Our work fills this gap by examining the environmental impact of cybersecurity infrastructure, including energy use of security systems, carbon emissions from threat detection processes, and environmental costs of cybersecurity hardware. We propose AI-driven solutions for energy optimization, waste reduction, and environmental impact management that address the projected 30-40% annual growth in AI/ML-driven services.
2) **Development of Unified Interdisciplinary Frameworks:** Existing literature treats green IT, AI-enabled cybersecurity, and environmental sustainability as separate domains [14], [21], [25]. Our paper applies and extends existing frameworks (Cybersecurity Maturity Model, SDGs) to create a single analytical framework that combines technical practices (hardware acceleration, model optimization) with sustainability goals. This goes beyond prior works that often focus only on technical solutions by providing clear classifications and interaction models for the social-technical and ethical aspects of green cybersecurity.
3) **Systematic Identification of Dual-Purpose AI/ML Applications:** While Roshanaei et al. [28] discuss AI's dual role in cybersecurity and various authors explore AI for sustainability separately, no complete identification exists of AI/ML techniques that improve both cybersecurity and environmental sustainability. We identify and categorize key areas including edge computing architectures, model optimization techniques (pruning, quantization), accelerator-based architectures, and predictive maintenance systems that serve both security and sustainability purposes.
4) **Critical Analysis of Implementation Challenges and Research Gaps:** Current research identifies specific challenges in AI adoption [27] or cybersecurity vulnerabilities [28] individually, but lacks complete analysis of challenges at the AI/ML-cybersecurity-sustainability intersection. Our work examines multiple limitations including high computational demands of AI security systems, ethical concerns (privacy, algorithmic transparency), integration complexity, and the trade-offs between security effectiveness and environmental costs. We provide a detailed classification of research gaps with priority rankings based on environmental impact potential.
5) **Proposing Holistic Interdisciplinary and Ethical Approaches:** While works by Uzondu and Lele [29] and Dede et al. [30] address cybersecurity's role in supporting SDGs, they lack complete frameworks for balancing security, sustainability, and social impact. Our paper emphasizes the importance of interdisciplinary collaboration and ethical guidelines that combine environmental stewardship with cybersecurity objectives. This adds a detailed social-technical perspective to prior works primarily centered on technological aspects, addressing the responsible adoption of AI in cybersecurity while protecting both environmental and security interests.
6) **Comprehensive Systematic Literature Review Methodology:** Using a structured review methodology with complete search strategy across multiple domains (AI/ML, cybersecurity, environmental sustainability), our work combines separated knowledge and identifies specific research gaps at the intersection of these fields. Unlike existing surveys that address these domains separately, we provide the first complete review that establishes "Green Cybersecurity" as a distinct research discipline with its own methodologies, evaluation criteria, and research agenda, serving as a foundational resource for future interdisciplinary studies.

II. REVIEW METHODOLOGY

The Review Methodology for this study is structured to systematically explore the intersection of green cybersecurity, artificial intelligence, and machine learning. The primary objective is to investigate the current state of green cybersecurity, assess the environmental impact of cybersecurity operations, and identify the challenges and opportunities associated with integrating AI and ML into sustainable cybersecurity practices. To achieve this, the study employs a comprehensive literature review approach, supported by a well-defined methodology. The method flowchart is shown in Figure 1 and a detailed list of research sources is listed in Table 1. We employed a systematic approach for the search and evaluation of research articles.

A. DATA COLLECTION

1) SYSTEMATIC INQUIRY AND DATA ACQUISITION

A comprehensive search was conducted on a variety of academic search engines and databases, as presented in Table 1. The sources encompass major search engines that enable the extraction of articles from different journals and conference proceedings. The research methodology flow chart is shown in Figure 1.

2) FORMATION OF SEARCH KEYWORDS AND SYNONYMS

The study used nine key search terms:

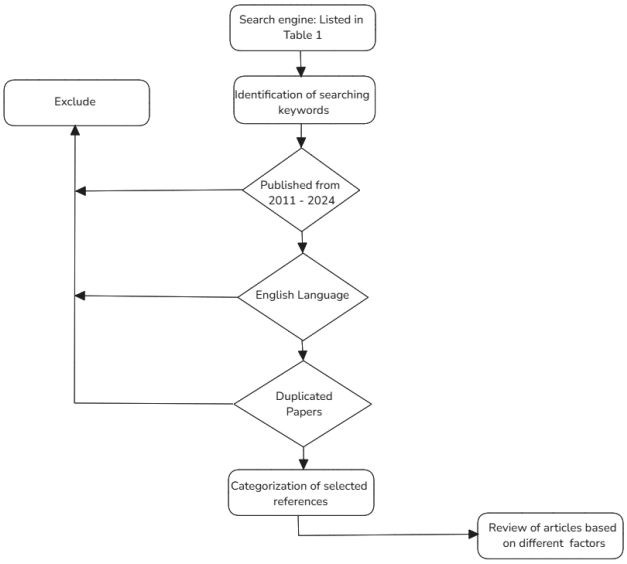


FIGURE 1. Research methodology flow chart.

TABLE 1. Articles searching database engines.

Search Engines	Database Address
ResearchGate	researchgate.net
Google Scholar	scholar.google.com
ScienceDirect	sciencedirect.com
IEEE Xplore	ieeexplore.ieee.org
MDPI	mdpi.com
EasyChair	easychair.org
Taylor & Francis Online	tandfonline.com
Journal of Reproducible Research	journalrrsite.com

- "Green Cybersecurity"
- "Artificial Intelligence"
- "Machine Learning"
- "Environmental Sustainability"
- "Large Language Models"
- "Threat Detection
- "Carbon Emissions"
- "Cybersecurity Maturity Framework"
- "Energy Optimization"
- "Sustainable Development Goals"

For each term, we have searched and included all available synonyms and related terms from various databases. By doing so, it was possible to build a very broad search string that could encompass all pertinent literature. Figure 2 presents the proportion of articles filtered by source, while Table 2 names the synonyms we considered applicable for each subject.

B. INCLUSION AND EXCLUSION CRITERIA

1) PUBLICATION DATE RANGE

We selected research articles published from 2011 to 2025. The time period was selected to ensure that our review would represent the most accurate state of current research

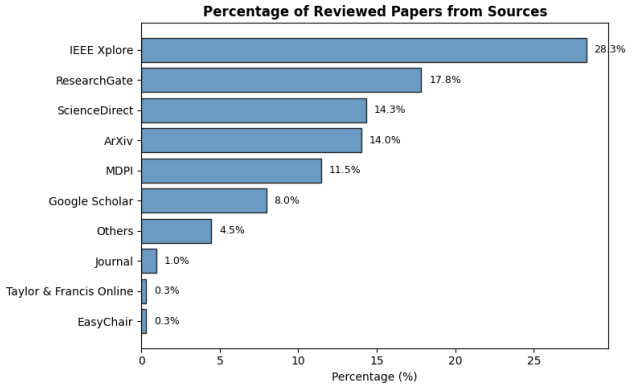


FIGURE 2. Percentage of reviewed papers from sources.

TABLE 2. Synonyms for search keywords.

ine **Keyword**	**Synonyms / Alternative Terms**
ine Green Cybersecurity	Eco-friendly cybersecurity, sustainable cybersecurity, environmentally conscious IT security
ine Artificial Intelligence	AI, machine intelligence, cognitive computing, smart automation
ine Machine Learning	ML, data-driven learning, algorithmic learning, statistical learning
ine Environmental Sustainability	Eco-sustainability, sustainable environment, green sustainability, resource conservation
ine Large Language Models	LLMs, big language models, advanced language models, neural language models
ine Threat Detection	Risk identification, intrusion detection, security monitoring, anomaly detection
ine Carbon Emissions	CO2 emissions, greenhouse gas emissions, carbon footprint, environmental emissions
ine Cybersecurity Maturity Model	Security maturity framework, cybersecurity capability model, cybersecurity readiness model
ine Energy Optimization	Energy efficiency, power optimization, energy management, resource optimization
ine Sustainable Development Goals	SDGs, global sustainability targets, sustainable objectives, UN sustainability goals
ine	

in green cybersecurity. In addition, priority was given to publications that were published after 2020 because these are expected to capture contemporary advances as well as trends in the field. Figure 3 illustrates the distribution of publication years, indicating the emphasis we have placed on the current literature.

2) INCLUSION AND RELEVANCE CRITERIA

We underwent an exhaustive filtering of the articles to identify only those that addressed the critical terms provided in Table 3. Each publication was screened for applicability within the field of green cybersecurity, with a focus placed on the environmental effects of cybersecurity operations and the

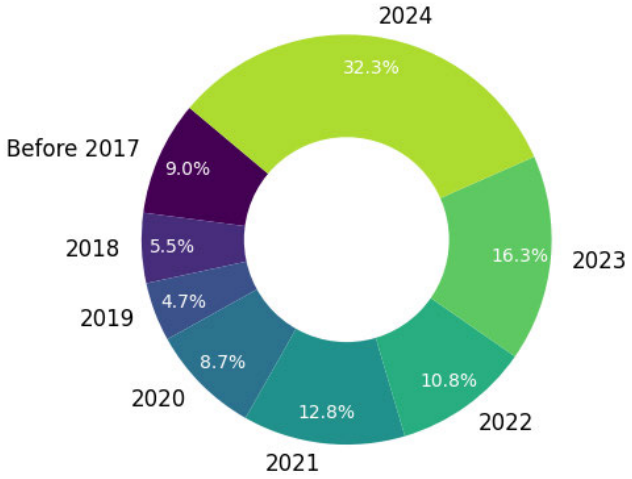


FIGURE 3. Percentage of papers included from 2011 to 2024.

incorporation of artificial intelligence, machine learning, and large language models. The selected articles had to address at least three key keywords (listed in Table 3) and explicitly discusses the relationship between cybersecurity operations and environmental sustainability.

III. THEORETICAL FRAMEWORK

The interconnection of cybersecurity and environmental sustainability is a relatively new concept that has gained increasing attention in recent years. As the global community becomes more aware of the environmental impact of digital technology, there is a growing recognition of the need to integrate sustainability into cybersecurity practices. This includes adopting energy-efficient technologies, optimizing resource utilization, and reducing the environmental impact of cyber threats [31]. Environmental sustainability has become a pressing global concern as human activities increasingly threaten the planet's ecological balance. It refers to the practice of using natural resources responsibly to ensure long-term environmental quality and human well-being, environmental sustainability aims to meet present needs without compromising ability of the future generations to meet theirs [32]. This involves the efficient use of resources, minimizing ecological footprints, and promoting long-term ecological balance. In the context of technology, environmental sustainability with efforts to reduce energy consumption, manage electronic waste, and limit carbon emissions.

Cybersecurity, according to Aslan et al. is the practice of protecting systems, networks, and data from digital attacks, unauthorized access, and data breaches [33], [34]. Cybersecurity guarantees that only authorized users have access to the information [35]. To improve protection, it is vital to understand the main types of cybersecurity. As the digital landscape expands, cybersecurity has become critical in safeguarding sensitive information and ensuring the integrity of digital infrastructures. However, increasing complexity and scale of cybersecurity measures, particularly in data-intensive operations, can have significant environmental impacts, such as increased energy consumption and electronic waste generation [36].

AI and ML are transformative technologies with significant potential to support cybersecurity and the achievement of sustainable development goals (SDGs) [37], [38]. Collins et al. defined it as the capability of machines to perform tasks that typically require human intelligence, such as decision-making, pattern recognition, and visual perception [39]. On the other hand, Dramsch et al. [40] refers to ML as a subset of AI that involves the development of algorithms that allow systems to automatically learn from data and improve their performance over time without being explicitly programmed. These technologies have been rapidly growing and have had substantial impacts in various decision-making domains, including healthcare, industry, agriculture, education, and finance [38], [41]. In addition, Vallabhaneni in his study states that the advent of artificial intelligence (AI) and machine learning (ML) technologies has revolutionized cybersecurity practices, offering novel solutions to combat the increasing complexity of cyber threats [26]. Furthermore, Eze et al. in their research state that the intersection of cybersecurity and sustainable development has garnered significant attention owing to the pivotal role of secure digital infrastructures in supporting economic growth, ensuring social inclusivity, and preserving environmental integrity [3]. As technology evolves, the increased digitization of various sectors has rendered cybersecurity an imperative facet of sustainable development endeavors [42].

A. INTERCONNECTION BETWEEN AI, ML, CYBERSECURITY, AND SUSTAINABILITY

AI and ML technologies have been increasingly applied to various sectors, including environmental sustainability and cybersecurity to address complex challenges. In the environmental sectors, AI and ML contribute to the analysis and prediction of ecological patterns and climate phenomena, while in cybersecurity, these technologies improve threat detection and response capabilities [47].

Integrating AI and machine learning into cybersecurity opens opportunities for environmental sustainability. For example, as explained by Evans and Gao, Google's DeepMind AI is a good example; it optimizes the cooling systems in Google's data centers, resulting in a 40% reduction in energy consumption, lower costs, and a smaller carbon footprint [48]. According to the Google Environmental Report (2024) later written in a blog by Google itself, Google further advanced its efficiency with its sixth generation Trillium TPU, which is over 67% more energy efficient than its predecessor. In addition, its data centers now average a power usage effectiveness (PUE) of 1.10, about 1.8 times more efficient than typical enterprise data centers [49]. In addition to hardware, Google has adopted methods that reduce the energy needed to train AI models by up to

TABLE 3. Literature review of green cybersecurity research.

Paper	Year	Green Cyber	AI	ML	Env. Sustain	LLMs	Threat Detect	Carbon Emiss	CMM	Energy Opt	SDGs	Our Paper Connection
[8]	2023	N/A	✓	N/A	✓	N/A	N/A	N/A	✓	✓	✓	Framework alignment
[9]	2023	N/A	✓	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	AI/ML foundation
[10]	2024	N/A	✓	✓	N/A	✓	✓	N/A	N/A	N/A	N/A	LLM testing frameworks
[11]	2024	✓	✓	✓	✓	N/A	N/A	N/A	N/A	✓	✓	Core theme address
[14]	2024	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	N/A	✓	SDG integration
[17]	2023	N/A	✓	✓	✓	N/A	N/A	N/A	N/A	N/A	✓	Environmental AI
[22]	2021	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	✓	N/A	Policy framework
[25]	2024	✓	N/A	N/A	✓	N/A	✓	✓	N/A	N/A	N/A	Resilience eng
[26]	2024	✓	✓	✓	N/A	N/A	✓	N/A	N/A	✓	N/A	Energy-aware security
[28]	2024	N/A	✓	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	ML architectures
[30]	2024	✓	N/A	N/A	✓	N/A	N/A	N/A	✓	N/A	✓	ESG compliance
[37]	2023	N/A	✓	✓	✓	N/A	✓	N/A	N/A	✓	N/A	Carbon impact
[60]	2024	✓	✓	✓	✓	N/A	✓	N/A	N/A	✓	N/A	IoT protocols
[72]	2024	N/A	✓	✓	N/A	✓	✓	N/A	N/A	✓	N/A	Spam detection
[76]	2020	N/A	✓	✓	✓	N/A	✓	N/A	N/A	✓	N/A	Climate alignment
[81]	2024	✓	✓	✓	✓	N/A	N/A	N/A	N/A	✓	N/A	Green computing
[98]	2022	✓	✓	✓	✓	✓	✓	✓	N/A	✓	✓	LLM ethics in security
[104]	2023	✓	✓	N/A	✓	N/A	✓	N/A	✓	N/A	N/A	Maturity models
[116]	2023	N/A	✓	✓	✓	N/A	✓	N/A	✓	N/A	N/A	Energy laws
[127]	2024	✓	✓	✓	✓	N/A	✓	N/A	✓	N/A	✓	Grid protection
[153]	2024	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	✓	N/A	Energy
[11]	2024	✓	✓	✓	✓	N/A	✓	N/A	✓	✓	✓	Green Technology Innovation
[54]	2021	✓	✓	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	Threat frameworks
[238]	2021	N/A	✓	✓	✓	N/A	✓	N/A	✓	N/A	N/A	Training optim
[245]	2023	N/A	✓	✓	✓	N/A	✓	N/A	N/A	✓	N/A	Emission tracking
[263]	2024	✓	N/A	N/A	✓	N/A	N/A	✓	✓	✓	✓	Metrics basis
[266]	2024	✓	✓	✓	✓	N/A	✓	N/A	N/A	N/A	N/A	Adversarial impacts
[272]	2024	✓	✓	✓	✓	N/A	✓	✓	N/A	✓	✓	Lifecycle approach
[292]	2018	✓	N/A	N/A	✓	N/A	N/A	N/A	N/A	✓	N/A	Blockchain energy
[313]	2024	✓	N/A	N/A	✓	N/A	N/A	N/A	✓	N/A	N/A	Quantum protection
[29]	2024	✓	✓	✓	✓	N/A	✓	N/A	N/A	✓	✓	IoT optimization
Our Paper	**2025**	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	**Green Cybersecurity**

100 times and cut related emissions by up to 1,000 times [49]. Google's AI tools also extend beyond its internal operations. For example, as further stated in the Google Environmental Report (2024), its fuel-efficient routing system has lowered greenhouse gas emissions by more than 2.9 million metric tons since its launch in late 2021, roughly the same as taking about 650,000 fuel-based cars off the road for a year [49]. In addition, Google's flood prediction models are used in more than 80 countries, providing early warnings to communities so they can prepare for river floods and protect over 460 million people [49]. Similarly, Microsoft's AI for Earth initiative leverages AI to improve energy efficiency in its data centers, supporting the company's goal of becoming carbon negative [50].

Unfortunately, AI and machine learning (ML) applications present challenges while offering significant benefits in both fields. For example, in cybersecurity, there is a need for ethical considerations and protection of data privacy when adopting AI and ML solutions [51], [52]. In addition, environmental applications must consider the precision and reliability of AI models in predicting natural systems, which are inherently complex and dynamic [53]. The intersection of AI and ML with environmental and cybersecurity is a promising area of research and application. Sarker et al. in their research state that the integration of AI and ML into cybersecurity has transformed the way digital threats are detected and managed, offering enhanced capabilities for real-time monitoring, automated threat response, and predictive analytics [54]. However, the deployment of these technologies must be carefully managed to address ethical concerns, data privacy issues, and the inherent complexities of natural and digital ecosystems. Table 4 illustrates Google's renewable energy consumption from 2019-2023 and Table 5 shows a breakdown of renewable energy for 2023.

While the systems theory model highlights the active interaction between technical systems and sustainability constraints, the SDG Framework situates cybersecurity within broader global ecological goals, and the CMM offers a maturity continuum for assessing both security and environmental performance. Together, these frameworks provide complementary perspectives that support systems-level modeling, policy alignment, and organizational benchmarking to integrate green principles into cybersecurity. Table 4 and Table 5 shows a breakdown of renewable energy from 2019 to 2023. Google consistently increased its renewable energy capacity, reaching more than **25 million MWh** of renewable electricity purchased by 2023 and maintaining **100% of renewable electricity usage** in all operations. The company's energy portfolio includes power purchase agreements (PPA), on-site generation, and grid-based renewables. In 2023 alone, Google's total energy consumption was approximately **25.9 million MWh**, with **61%**

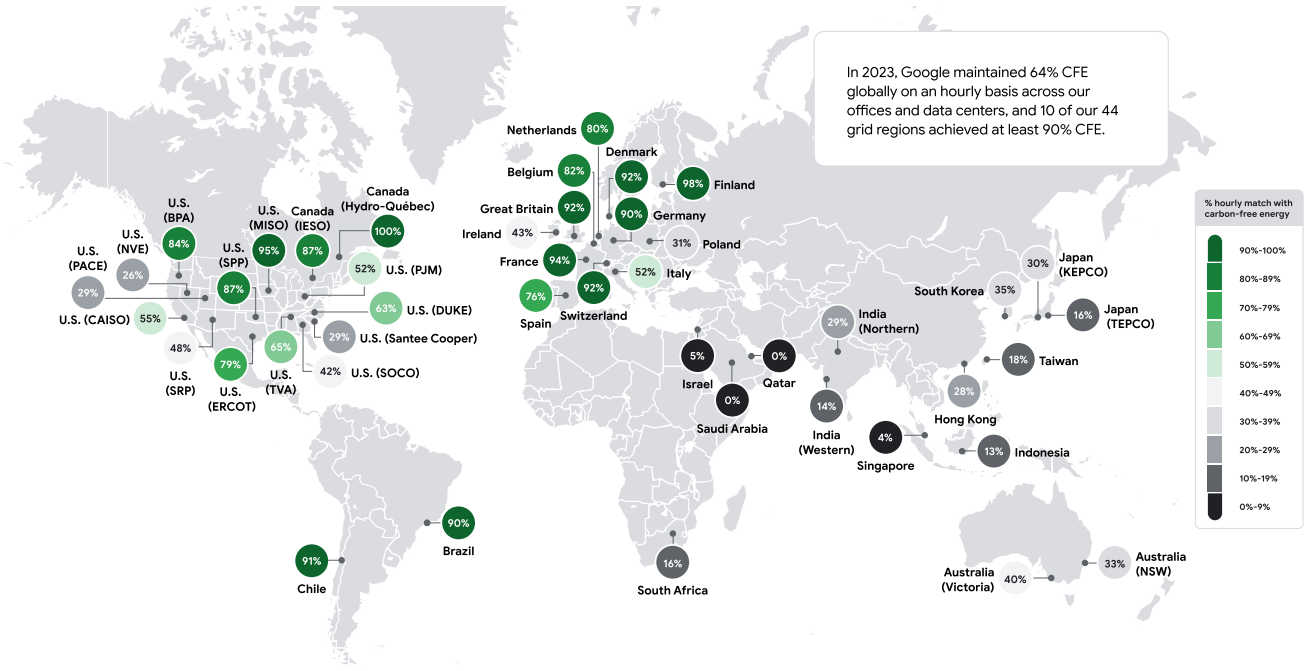


FIGURE 4. Google's global data center carbon-free energy map [64].

TABLE 4. Google's renewable energy consumption and contracts (2019–2023) [64].

Energy Consumption	Unit	2019	2020	2021	2022	2023
Renewable energy contracts (cumulative)	MWh	5,400	5,700	7,200	11,600	14,900
Renewable electricity (PPAs)	MWh	9,715,000	12,069,200	14,109,400	16,693,600	19,089,200
Renewable electricity (on-site)	MWh	6,300	7,200	8,800	9,600	10,700
Renewable electricity (grid)	MWh	2,515,900	3,062,100	4,168,900	5,073,000	6,207,100
Total renewable electricity purchased	MWh	12,237,200	15,138,500	18,287,100	21,776,200	25,307,000
Electricity purchased from renewable sources	%	100	100	100	100	100

sourced from renewables and **39% from non-renewable sources**, primarily driven by purchased electricity. This reflects Google's significant strides toward a carbon-free and sustainable digital infrastructure.

IV. THEORETICAL MODELS EXPLAINING THE INTEGRATION OF AI AND ML INTO CYBERSECURITY ENSURING SUSTAINABILITY OF THE ENVIRONMENT

To formulate theoretical models that clarify the integration of AI and ML in cybersecurity while maintaining environmental sustainability, several frameworks that cover the technological, ecological, and ethical aspects of this integration can be taken into account [55]. These frameworks describe the essential cybersecurity procedures to protect against any potential threats in addition to offering the structural framework for implementing and overseeing cybersecurity measures while advancing sustainable practices. We have discussed some of the popular models and frameworks.

A. SYSTEMS THEORY MODEL

The Systems Theory Model provides a comprehensive structure for understanding the incorporation of AI and ML into cybersecurity and ensuring environmental sustainability [56]. This model highlights the relationship between technology, environmental variables and societal implications, allowing for a comprehensive understanding of how these aspects interact, increasing cybersecurity measures sustainably [57]. Systems theory proposes that complex systems are formed of interconnected components that work together to attain a shared objective. In the context of cybersecurity, this model helps to examine how AI and ML could be incorporated into current security frameworks while addressing environmental sustainability [22]. When applied specifically to AI-driven cybersecurity solutions, Systems Theory illuminates how machine learning algorithms can be optimized to detect threats while minimizing computational resource usage, thereby reducing energy

TABLE 5. Google's energy consumption breakdown for 2023 [64].

Energy Consumption Unit (MWh)	Renewable Sources (MWh)	Non-Renewable Sources (MWh)	Total (MWh)
Fuel	68,300	232,900	301,200
Purchased electricity[7]	15,672,000	9,580,600	25,252,600
Purchased heat[8]	0	278,500	278,500
Purchased steam	0	14,500	14,500
Purchased cooling	0	53,000	53,000
On-site renewable electricity	10,700	0	10,700
Total energy consumption	15,750,900	10,159,600	25,910,500

Notes: [7]Purchased electricity figures represent contracted energy.
[8]Purchased heat figures represent energy acquired as heat.

consumption and carbon footprint compared to traditional security methods [56]. Reference [58] proposes several approaches of systems theory including the Value-Driven Approach, Socio-technical Approach, Project-Management Approach, and Knowledge-Management Approach. The sociotechnical method is considered appropriate for tackling cybersecurity issues due to its incorporation of technological, organizational, and human components [59]. Within this approach, AI and ML serve as technological components that adaptively respond to emerging threats, while environmental sustainability considerations represent critical social and organizational values that must be balanced with security objectives. For example, AI-powered anomaly detection systems can identify security breaches using fewer computational resources than rule-based systems that require constant updating and extensive database searches, demonstrating how technological advancement through AI directly contributes to environmental sustainability [60]. This model can be represented by the equation.

$$S = f(T, E, S) \tag{1}$$

This equation demonstrates that system performance (S) is a function of technology (T), environmental factors (E), and social factors (S). By optimizing each component, organizations can improve their cybersecurity posture while promoting sustainability.

B. SUSTAINABLE DEVELOPMENT GOALS (SDG) FRAMEWORK

The SDG Framework is an internationally acknowledged structure that matches technical innovations, including AI and ML, with sustainability aims, rendering it an optimal perspective for assessing their incorporation into cybersecurity and preserving environmental sustainability [61]. The SDGs, instituted by the United Nations, offer a comprehensive framework for achieving a sustainable future. According to [62], there are two aspects of sustainable AI: the sustainability of AI, which addresses the effects of developing and applying AI technology on the environment, and the AI for sustainability, which focuses on applying AI to accomplish the SDGs. The incorporation of AI and ML into cybersecurity may be matched with these objectives to guarantee that technology progress fosters environmental sustainability and societal welfare. AI-powered threat detection systems that utilize efficient algorithms require significantly less energy than traditional security operations centers staffed 24/7, directly contributing to reduced carbon emissions while maintaining robust security postures [63]. This integration can be modeled as:

$$SDG_i = \sum_{j=1}^{n} (C_j R_j) \quad \ldots\ldots\ldots \text{(Equation II)} \tag{2}$$

where:

- SDG_i = Contribution to the ith Sustainable Development Goal.
- n = Total number of cybersecurity measures considered.
- C_j = Cybersecurity measure j, such as:
  - ML-based vulnerability management,
  - AI-driven threat detection,
  - LLM-powered compliance automation.
- R_j = Resource efficiency of measure j, accounting for:
  - Environmental impact,
  - Computational efficiency,
  - Energy consumption.

Equation (2) models the integrated contribution of cybersecurity measures toward SDGs, emphasizing that the positive environmental impact depends on both the effectiveness of the measures and their resource efficiency. It provides a framework for evaluating and optimizing cybersecurity practices in the context of sustainability.

1) RELEVANT SDGS FOR AI AND ML IN CYBERSECURITY

Several SDGs align with the integration of AI and ML in cybersecurity, particularly those that focus on sustainability, innovation, and infrastructure [65]:

- SDG 9 Industry, Innovation, and Infrastructure

- SDG 11 Sustainable Cities and Communities
- SDG 12 Responsible Consumption and Production
- SDG 13 Climate Action

Several SDGs align with the incorporation of AI and ML into cybersecurity, especially those that address environmental sustainability, innovation, and infrastructure [65]. For example, according to SDG 9, ML-driven anomaly detection systems promote resilient infrastructure by identifying potential cyber threats to critical systems while consuming fewer computational resources than traditional signature-based approaches [66], [67]). According to SDG 12, AI optimization algorithms enable responsible consumption by dynamically allocating cybersecurity resources based on real-time threat assessments, preventing overprovisioning of computing power and reducing overall energy consumption [68]. These practical applications demonstrate how AI and machine learning improve cybersecurity capabilities and also advance sustainability objectives. The authors in [22] highlight the essential importance of acknowledging the interconnection between cybersecurity and sustainable development. By coordinating cybersecurity initiatives with the SDGs, society can use the transformational power of digital technology to establish a safe, inclusive, and sustainable future for everyone [58].

C. CYBERSECURITY MATURITY MODEL (CMM)

The Cybersecurity Maturity Model (CMM) was developed to measure and enhance an organization's cybersecurity capabilities [53]. It presents a framework for incorporating new technologies like AI, ML, and LLM into cybersecurity processes while addressing environmental sustainability. The CMM integrates environmental sustainability by measuring the energy efficiency and resource optimization of cybersecurity procedures, particularly focusing on how AI and ML implementations can be optimized for both security effectiveness and environmental impact. For example, AI-powered security systems can be designed to operate more efficiently during low-threat periods, automatically scaling their resource consumption based on actual security needs, thereby reducing unnecessary energy usage while maintaining robust protection.

The CMM is commonly used in organizations that handle sensitive information, including military, healthcare, and finance, to verify compliance with security requirements and increase resilience against cyberattacks. In these sectors, the integration of AI and machine learning technologies has become crucial to maintaining security and achieving sustainability goals. The model can be represented by the following equation.

$$M = \frac{E + R + T \sum (\text{Implemented Practices})}{3 \times (\text{Total Required Practices})} \quad (3)$$

where

- M: Maturity level of cybersecurity techniques.
- E: Effectiveness of environmental consideration (e.g., energy efficiency, reduction of carbon footprint).
- R: Efficiency in resource management (e.g., optimal utilization of computational resources).
- T: Technology integration level (e.g., adoption of AI, ML, and LLM).

Each component (E, R, T) is evaluated on a scale of 0 to 1 (0% to 100% maturity), where 1 reflects the highest maturity level.

For example, using energy-efficient AI algorithms, optimizing data center operations to reduce carbon footprints, and promoting the use of renewable energy in cybersecurity infrastructure contribute to higher maturity levels. [69] describes some AI and ML applications that indicate their capabilities and maturity levels, helping organizations determine their current maturity level, as shown in (Table 6).

V. ENVIRONMENTAL IMPACT OF CYBERSECURITY

The environmental impact of cybersecurity has been a critical concern as the digital economy expands and the need to protect information systems increases. Cybersecurity operations are crucial for protecting data and infrastructure, but unfortunately consume a large amount of energy and emit carbon dioxide [70]. Understanding these impacts is important for establishing a method that improves both security and promotes global sustainability goals [71].

A. ENERGY CONSUMPTION

Energy consumption in cybersecurity operations is a critical concern, particularly as these systems expand. From research, cybersecurity operations are known to be highly dependent on energy-intensive infrastructure, especially data centers and network equipment [72]. According to a report from the International Energy Agency (IEA), data centers worldwide consumed about 1% of global electricity in 2020, a rate that is likely to increase as the digital transformation advances (IEA- 2023). However, the energy required to power these data centers directly adds to greenhouse gas emissions, especially if electricity is generated with fossil fuels [37]. In addition, as cybersecurity measures become more advanced and the volume of data increases, the need for computational power and energy increases [73]. The computational demands for processing large datasets and training sophisticated algorithms require substantial amounts of electricity [74]. Strubell et al. in their studies examine the energy consumption associated with the training of large deep learning models for Natural Language Processing (NLP) and discovered high carbon emissions resulting from extensive computational requirements, especially models such as GPT-3 and BERT. And we conclude that GPT-3 can consume up to 200 megawatt hours (MWh) of energy, equivalent to the annual electricity consumption of approximately 6-7 US households [75]. However, research by Nishant et al. indicates that AI techniques significantly enhance energy consumption analysis by identifying patterns and optimizing efficiency opportunities in energy management systems [76].

TABLE 6. AI and ML applications indicating their capabilities and maturity levels for organizations.

S/N	Application	AI & ML Capabilities	Maturity Level	Remarks	References
1	Threat Prevention and Detection	Pattern Recognition, Classification, Clustering	**High**	Implements strategies and tools to safeguard an organization's assets from threats by identifying harmful contents that compromise integrity, privacy, or availability.	[78], [79], [80]
2	Endpoint and Cloud Security	Pattern Recognition, Classification, Clustering, Anomaly Detection, Ranking	**High**	Monitors and examines activities on devices and servers, such as network connections, file operations, and user actions, to detect and mitigate threats in real-time.	[81], [82], [83]
3	User and Entity Behaviour Analytics (UEBA)	Behavioural Analysis, Clustering	**Medium**	Evaluates user, application, device, and entity behavior within an organization to identify security threats and anomalies. Useful for detecting insider threats and compromised accounts.	[84], [85], [86]
4	Network Security	Classification, Anomaly Detection, Pattern Recognition	**Medium**	Monitors and analyzes large network traffic volumes across complex infrastructures. Detects anomalies and suspicious activities indicating unauthorized access or potential attacks.	[87], [88], [89]
5	Security Analytics (SIEM)	Pattern Recognition, Anomaly Detection, Information Retrieval, Ranking, Generation	**Medium**	Provides a holistic view of security posture, facilitating identification and mitigation of emerging threats.	[90], [91]
6	Threat Intelligence	Classification, Clustering, Information Retrieval, Generation	**Low**	Collects, processes, and analyzes data to understand threat actors' goals, targets, and attack strategies to anticipate and prevent cyberattacks.	[92], [93], [94], [95]
7	Compliance and Risk Management	Classification, Information Retrieval, Ranking, Generation	**Low**	Identifies, estimates, and prioritizes cybersecurity risks associated with operations, assets, and individuals.	[96], [97], [98]
8	Vulnerability Management	Classification, Pattern Recognition, Ranking, Generation	**Low**	Identifies, assesses, and mitigates security flaws in IT systems. Supported by public vulnerability databases like the CVE list.	[99], [100], [101]

B. CARBON FOOTPRINT OF CYBER THREATS

The carbon footprint of cyber threats has become an area of concern as digital growth increases. Cybercrime not only incurs economic costs, but also contributes to carbon emissions through increased energy consumption in data centers and digital infrastructures [77]. For example, cyberattacks such as malware, botnets, and distributed denial-of-service (DDoS) have significant environmental impacts, particularly those with large incidents. Ransomware has also started to target critical infrastructure and cyber-physical systems (CPS), such as industrial control systems, smart grids, and healthcare networks [5].

Note: The "-" indicates data was not available or not reported for the respective year.

1) STRATEGIES FOR REDUCING ENERGY CONSUMPTION AND CARBON FOOTPRINT

It was stated by Kumawat that one of the approaches to mitigating energy consumption is through model optimization techniques, such as pruning, quantization and knowledge distillation [102]. Pruning involves removing redundant parameters from ML models to reduce their complexity and computational demands [103]. Quantization reduces the precision of computations, thus lowering the energy required for modeling during training and inference [104]. In the research of Kumawat deduce that pruning can lead to significant energy savings while maintaining model accuracy, and concluded that model pruning can reduce the number of parameters in a neural network by up to 90%, which significantly decreases energy usage [105]. Building on earlier research by Kumawat [105], which showed that pruning can remove up to 90% of the parameters of a neural network without losing accuracy or wasting energy, newer studies highlight how pruning helps the environment. For example, Han et al. [106] found that simplifying models such as AlexNet and VGG-16 through unstructured pruning cuts energy use by 3 to 7 times precision intact. Structured pruning goes even further: The study by Mugnaini et al. states

TABLE 7. Carbon-free energy (CFE) metrics (Hourly) for google facilities (2019–2023).

Metric	Unit	2019	2020	2021	2022	2023
CFE across Google data centers (hourly)	%	61	67	66	64	64
CFE across Google offices (hourly)	%	-	-	-	541	56
CFE across Google data centers and offices (hourly)	%	-	-	-	64	64

that pruning reduced energy use by 66.Several studies have demonstrated the effectiveness of pruning and other optimization techniques in reducing energy consumption and carbon emissions in deep learning. For instance, one reported energy savings of 99% and a reduction in carbon emissions by 68.75% [107]. Petri et al. achieved a 73.9% reduction in power usage with only a minor decrease in accuracy [108]. Similarly, Reguero et al. showed that pruning could actually improve model performance–reducing energy consumption by 56.5% while increasing validation accuracy by 2.38% [109]. Widmann et al. reported balanced results, achieving 59.06% energy savings with a minimal accuracy drop of 3.01% [110], while You et al. demonstrated that pruning scales effectively, achieving 4.7× energy savings without degrading performance [111]. Additionally, Pirnat et al. found that pruning can reduce carbon footprints by 42% compared to more advanced models [112]. These optimization techniques are summarized in Figure 6, while Figure 5 illustrates the process flow for network pruning, knowledge distillation, and quantization.

Hubara et al. explored the use of low-precision arithmetic in Deep Neural Networks (DNNs) to reduce computational and energy demands. They trained Quantized Neural Networks (QNNs) on MNIST, CIFAR-10, SVHN, and ImageNet, and found that QNNs maintained predictive performance comparable to their 32-bit counterparts [113]. Furthermore, Desislavov et al. emphasized the importance of selecting energy-efficient hardware–such as GPUs optimized for lower power usage–as a strategy to energy requirements during training and inference [114]. Salehin et al. also concluded that model compression and AutoML frameworks can improve deep learning efficiency, especially for resource-constrained devices [115].

For energy and security effectiveness verification and monitoring of AI-augmented systems, there are measurable metrics accessible through more recent benchmarking studies. Han et al. showed that structured pruning reduced energy usage by 66.99% with zero loss in detection performance [196]. Analogously, Jouppi et al. benchmarked Google TPUs with existing CPUs/GPUs and recorded 15-30× speedup and 30-80× energy efficiency for DNN inference. Relative analysis by Widmann et al. resulted in 59.06% energy reduction with only 3.01% drop in classification performance [199]. These performance metrics serve for comparative analysis as a basis for green cybersecurity design trade-off monitoring.

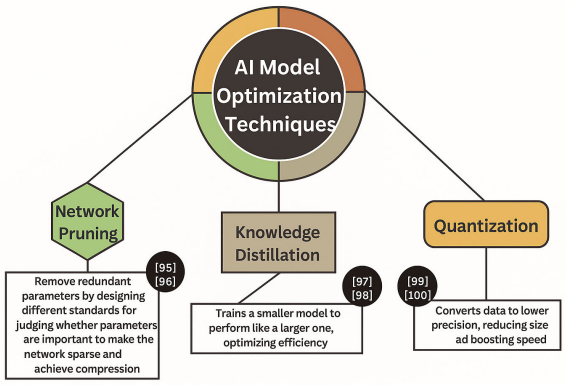


FIGURE 5. AI model optimization techniques.

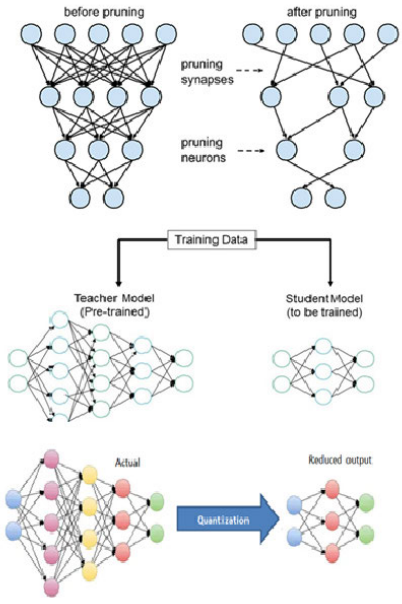


FIGURE 6. AI model optimization techniques process a. Network Pruning b. Knowledge c. Quantization [117], [118].

Recent research highlights several innovative approaches; for example, research by Eleftheriadis C et al. presents a new energy-efficient Fast Fourier Transform (FFT) architecture

that helps in the reduction of required memory and improves energy efficiency [119]. In addition, Zhong et al. use algorithms such as sparsity to reduce the memory footprint of regular networks to fit mobile devices and to shorten training time for networks [118]. Moreso, Usman et al. propose deep learning algorithms like YOLOv5 and Faster R-CNN for effective object detection in crime surveillance, demonstrating high accuracy in real-time accuracy in real-time applications [94]. Chataut et al., on the other hand, focus on developing optimization algorithms that improve efficiency and accuracy for various ML tasks, including handling noisy functions [72]. Additionally, Tramm J et al. demonstrate significant performance improvements in Monte Carlo simulations on AI accelerator hardware, emphasizing the importance of algorithm optimization in optimization in high-performance computing [120].

2) RENEWABLE ENERGY SOURCES

Data centers are the major consumers of electricity and powering them with renewable energy sources can reduce both environmental impact and operational costs. For example, Goiri et al. developed the Parasol prototype, a green data center, and the GreenSwitch model, which manages workloads and energy sources to achieve cost improved efficiency [121]. In a similar vein, Dikaiakos et al. converted an academic data center into a solar powered facility using a cyber-physical system for continuous monitoring and configuration [122]. Li et al. also showed that metrics such as renewable power utilization and load tuning frequency are essential to designing sustainable data centers [123]. These examples demonstrate how renewable energy in data centers can reduce emissions while keeping operating expenses in check. Table 7 illustrates Google's carbon-free energy metrics for its facilities, while Figure 4 depicts Google's global data center carbon-free map.

In renewable energy sources, transitioning data centers is one of the most impactful strategies. According to Eleftheriadis et al. in their research, traditional data centers often use inefficient cooling systems; the improvement in this cooling technology, such as free air cooling and liquid cooling, has shown a reduction in energy consumption [124]. Hoefler et al. in their research highlighted how DVFS can lead to significant energy savings in a high-performance computing environment [113]. However, [125] and [126] studies emphasize the application of various ML algorithms, such as Random Forest, Support Vector Machines, and Decision Trees, to accurately estimate energy production from renewable sources such as solar and wind, effectively addressing their previously unpredictable behavior.

Beyond data centers, the integration of renewable energy can also improve the sustainability of other critical cybersecurity infrastructures [127]. Modern network equipment, such as routers, switches and firewalls, as well as security operations centers, require a constant and reliable power supply to ensure uninterrupted connectivity and robust monitoring [128]. By powering these elements with renewable energy, organizations can reduce their carbon footprint and also achieve cost savings similar to those seen in data centers [121], [129]. This holistic approach helps to create a more resilient cybersecurity ecosystem where both data processing and network defense benefit from sustainable energy practices [130].

3) CHALLENGES IN INTEGRATING RENEWABLE ENERGY

1) Energy Storage and Supply Variability: Solar and wind power are inherently intermittent. This means that their energy production can fluctuate due to weather conditions, seasonal changes, or time of day. For example, solar panels produce less energy on cloudy days or during the night. Such variability can lead to performance degradation and frequent tuning activities to match the energy supply with the demand, as highlighted by Li et al. Several research groups have proposed intelligent ways to address this challenge [123]. For example, an intelligent EMS framework proposed by Bhardwaj et al. dynamically balances energy loads among renewable sources (solar and wind), battery storage, and grid supply, optimizing consumption while minimizing grid dependency [131] and REDUX, a renewable energy manager, orchestrates distributed UPS devices to allocate energy resources efficiently, thus reducing costs and improving renewable energy utilization in data centers [132], [133].
2) Sustainable Practices in Data Centers: Maintaining a consistent balance between the energy generated from renewable sources and the consumption requirements of data centers and cybersecurity equipment is crucial to system stability. Achieving this balance often requires sophisticated management techniques. Demand side management (DSM) is an important method of balancing power supply and demand in electric grids, especially as more renewable energy sources are used. DSM helps consumers and utility companies make smart choices about how and when they use electricity, changing load patterns, and lowering peak demand [134]. Because renewable sources such as wind and solar produce electricity intermittently, the balance of supply and demand becomes more challenging as their use increases [135]. DSM offers a solution by providing incentives that encourage users to change their power use habits, which helps reduce peak loads and keep costs down [136]. This method is especially useful in smart grids, where the integration of renewable energy has made the balance of supply and demand more complex. Although energy storage systems are another option, their high costs often make DSM a more attractive choice for utilities around the world [136].
3) Regulatory and Ethical Considerations: Integrating renewable energy with smart grids presents several regulatory and ethical challenges. Regulators must

address issues such as disincentives for utilities, pricing inefficiencies, and cybersecurity risks [137]. In addition to the need for advanced technology, overcoming regulatory barriers through market liberalization and dynamic pricing is essential. There are also ethical concerns when using advanced tools such as artificial intelligence and machine learning in energy systems, which require clear guidelines for transparency, fairness, and accountability [138]. The idea of embedded energy storage adds more complexity, raising questions about ownership, compensation, and performance standards [139]. Moreover, the development of standardized protocols for emergency situations, such as black-start processes, remains a critical gap in current research [140]. As the use of renewable energy increases, regulations must also evolve to ensure that grids remain flexible, reliable, and sustainable [137], [140].

4) OPPORTUNITIES FOR A GREENER CYBERSECURITY INFRASTRUCTURE

1) Hybrid Renewable Energy Systems (HRES): Hybrid Renewable Energy Systems are systems that combine more than one renewable source [141]. For example, solar and wind. This helps them overcome some setbacks. When one source performs poorly (for example, solar panels during a cloudy day), another (such as wind turbines) may still provide sufficient power [15], [141]. These systems are more efficient and cost-effective and produce lower carbon emissions compared to systems that rely on one single source [141], [142]. By making the grid more stable and meeting growing energy needs in a sustainable way, HRES offer new opportunities for a greener cybersecurity infrastructure. This ensures that essential systems continue to operate safely while reducing their environmental impact.
2) Advanced Energy Management Technologies: Big data analytics and machine learning are essential to improve energy management. They allow for better prediction and optimization of energy use and resources [143]. Machine learning models, such as random forests and neural networks, have demonstrated superior performance in predicting energy consumption in various sectors [144]. The integration of big data with the Internet of Things (IoT) platforms further enhances energy management capabilities, allowing the development of innovative home energy management systems [145]. These advanced technologies enable efficient electricity generation forecasting, and some systems achieve prediction accuracy close to 99% of actual usage [146].
3) Cyber-physical systems (CPS): Cyber-physical systems (CPS) in data centers allow for the combined management of computer systems and physical equipment, improving both performance and energy efficiency. These systems link IT operations with mechanical components, creating a unified control strategy that balances application performance and energy use [147], [148]. In integrated energy systems, CPS improves the operation of various energy subsystems, although it may also raise cybersecurity risks [149]. To address these risks, deep reinforcement learning is integrated into the MAPE-K control loop to improve anomaly detection and enable adaptive decision-making [150]. This integration strengthens security and also optimizes the use of energy in CPS.

5) AI AND ML FOR GREEN CYBERSECURITY

AI-driven predictive analytics can improve cybersecurity efficiency by addressing potential threats and optimizing resource allocation. Lifecycle assessments (LCA) also improve the evaluation of environmental impacts throughout the life cycle of renewable energy systems, enabling real-time monitoring and optimization of resource use [157]. Furthermore, AI and ML facilitate advances in biomass and biofuels by optimizing feedstock management and improving combustion systems, leading to reduced emissions and improved fuel efficiency, Overall, synergy between AI, ML and renewable energy not only supports the transition to a low carbon economy, but also fosters strategic decision-making for sustainable energy deployment [158].

6) RESOURCE UTILIZATION

The development and deployment of AI and ML models in cybersecurity presents significant opportunities to optimize resource utilization while addressing environmental impacts [9]. Research indicates that while AI technologies improve cybersecurity by improving accuracy and scalability, they also come with significant resource demands, including computational power, data storage, human expertise, energy consumption, raising environmental concerns [159], [160]. From research, training of AI and ML models, especially those with deep learning architectures, requires substantial computational resources [161]. AI and ML training process can last for days or even weeks, consuming huge amounts of energy, and the model also requires large datasets for training and validation, leading to an increased demand for data storage solutions. In the [162] finding, it was stated that most of the time advanced models such as deep neural networks (DNN) and the large language model (LLMs) require massive datasets, in general, the development of the AI and ML model requires specialized expertise in data science, machine learning, and domain-specific knowledge [41].

7) SPECIALIZED ML ACCELERATORS

Emerging hardware accelerators, namely Tensor Processing Units (TPUs), Graphcore Intelligence Processing Units (IPUs), and Application-specific Integrated Circuit (ASIC) encryption chips, are transforming cyber security operations through energy efficiency [163]. Accelerators improve

TABLE 8. Comparison of ML models in energy applications: Typical accuracy, energy overhead, and key applications.

ML Model	Typical Accuracy	Energy Overhead	Key Applications	References
Random Forest	85–95%	Medium	Load forecasting, anomaly detection	[151]
Neural Networks	90–99%	High	Demand prediction, generation forecasting	[152]
Gradient Boosting Machines (GBM)	88–96%	Medium–High	Energy consumption optimization	[153]
Support Vector Machines (SVM)	80–92%	Medium	Short-term load forecasting	[154], [155]
Linear Regression	75–85%	Low	Baseline energy usage prediction	[156]

compute-intensive workloads such as machine learning (ML) inference, cryptographic functions, and real-time threat detection while significantly lowering power consumption against conventional CPUs and GPUs [164]. For example, as stated by Jouppi in his research, the first-generation tensor processing unit (TPU) is 15-30 times faster with 30-80 times higher energy efficiency for deep neural network (DNN) inference compared to current CPUs and GPUs in similar semiconductor technologies [165]. In addition, according to Pandey, Google's Tensor Processing Unit (TPU), Deep Neural Network (DNN) accelerator, has replaced CPUs/GPUs in its data centers, with more than 15 × rate of inference [166]. Similarly, ASIC-specific hardware for encryption (such as AES-256 acceleration) reduces power consumption, with less power consumed for secure data transfer, a foundation of AES-based cybersecurity systems [167].

The integration of accelerators such as vision processing units (VPUs) and field programming gate arrays (FPGAs) into the design of the data center infrastructure makes the infrastructure much more energy efficient [168]. Research by Segio Rivas Gomez et al. explores the Vision Processing Unit (VPU), which is described as a highly parallel unit for vector processing that works within a power budget below 1W. In their research, they discovered that a multi-VPU configuration provides similar performance relative to baseline CPU and GPU architectures, but thermal design power (TDP) is reduced by up to 8x compared to them [169]. FPGAs, which are dynamically programmable, reconfigure their logic to stay ahead of new threats, such as zero-day attacks, while maintaining moderate energy consumption [170]. For cloud-based cybersecurity workloads, secure execution environments like AccShield enable secure and low-latency shared accelerator access for cloud-based cybersecurity workloads without avoiding vulnerabilities such as side channel attacks at the cost of an energy-saving trade-off [171].

But embracing those accelerators involves difficult trade-offs. ASICs, though unbeatable inefficiency for dedicated tasks such as encryption, do not put the flexibility of programmable solutions like FPGAs. At the edge - such as in IoT sensors or network gateways - the energy budget is tight, and so hardware and DNN must be codesigned to deliver accuracy to meet a system-appropriate level of performance [172]. Emerging technologies, such as neuromorphic accelerators for spiking neural networks, have the potential for higher efficiency by emulating biological computational processes, thus facilitating adaptive threat learning without high power consumption [173]. These kinds of technological innovation are in line with international sustainability aspirations (e.g., SDG 9 and 13), demonstrating how hardware can reconcile intensive cybersecurity measures with a lower environmental footprint.

8) GREEN ARTIFICIAL INTELLIGENCE

To mitigate these impacts, the initiative of promoting "green artificial intelligence" has emerged, focusing on the development of high-performance AI systems while minimizing energy consumption, model efficiency, and sustainability [174]. In addition, integrating robust cybersecurity measures with green technologies is essential to balance sustainability goals with security needs. This integration can lead to innovative approaches that improve both environmental and cybersecurity outcomes [175]. The exploration of interdisciplinary strategies, including adaptive cybersecurity frameworks and the use of emerging technologies like blockchain; can further optimize resource use in smart environmental applications. Thus, a concerted effort to develop ethical norms and innovative defense mechanisms is crucial to advance green cybersecurity [176].

9) LIFECYCLE ASSESSMENT (LCA)

The lifecycle impacts of AI and ML technologies cover various stages, from development to deployment and maintenance [177]. LCA has become significant due to the potential to enhance cybersecurity while minimizing environmental impact, but they also present challenges that need to be addressed [178]. The challenges include hardware used for AI and ML acceleration in cybersecurity applications that can face reliability issues due to device-level non-idealities and circuit-level faults, which could affect overall system performance and life span [11]. In addition, during the development phase, AI and ML models require substantial computational resources for training, which can lead to high energy consumption and associated carbon emissions [29].

This is especially important for complex models used in cybersecurity applications, such as anomaly detection and threat intelligence [179]. However, the long-term benefits of these technologies in improving energy efficiency and reducing the overall environmental impact can outweigh the initial investment in resources [180].

C. TRADE OFFS BETWEEN CYBERSECURITY EFFECTIVENESS AND ENVIRONMENTAL SUSTAINABILITY

The intersection of cybersecurity and environmental sustainability presents complex challenges that require careful consideration of competing priorities. Extensive research in this field reveals several critical trade-offs that organizations must navigate when implementing green cybersecurity solutions powered by artificial intelligence and machine learning technologies.

1) COMPUTATIONAL OVERHEAD VS. ACCURACY TRADE-OFFS IN THREAT DETECTION SYSTEMS

Neural network-based threat detection systems face a critical balance between achieving high accuracy and minimizing computational costs. Traditional architectures such as Multi-Layer Perceptrons (MLPs) and Convolutional Neural Networks (CNNs) demonstrate robust accuracy (e.g. 99% for MLPs, 100% on benchmark datasets such as MSTAR for CNNs), but incur significant computational overhead, with classification times ranging from 10^3 to 1.5×10^4 seconds [181], [182]. For example, while CNNs excel in structured data (for example, 96. 88% accuracy on the SAMPLE dataset, their performance degrades in distributed or complex threat patterns, highlighting a trade-off between baseline accuracy and adaptability [183]. In contrast, hybrid and ensemble approaches, such as stacked ensembles that combine recurring neural networks (RNNs) with CNNs, or hybrids such as CNN-BiLSTM, demonstrate even higher accuracies (up to 99.9% or higher) [184] and, in some cases, can achieve detection times as low as 0.32 seconds. However, these advanced models typically incur steeper computational costs because of their complexity. For example, a CNN-BiLSTM model achieved 99. 89% accuracy in NSL-KDD and 98. 95% in UNSW-NB15, but required more intensive processing than traditional methods [184].

The trade-off between computational efficiency and detection accuracy, as explained in Table 9 is crucial, especially when considering the environmental impact of deploying these systems on a scale. The high computational overhead increases energy consumption and also contributes to a larger carbon footprint. To mitigate these trade-offs, optimized implementations leverage strategies like computation offloading and hardware acceleration [182], [185]. For instance, offloading detection tasks in vehicle intrusion systems reduced latency significantly, while neuromorphic hardware running Liquid State Machines cut power consumption [182], [186]. Similarly, data processing techniques such as Principal Component Analysis (PCA) and under sampling improved processing speeds at the cost of marginal accuracy losses (e.g., 3.01% decline for a 59.06% energy reduction) [184]. In particular, specialized ensembles achieved a 16.6× overhead reduction in malware detection on hardware without sacrificing accuracy, demonstrating that careful architectural design can reconcile efficiency and performance [187]. Table 8 illustrates the comparison of ML models in energy applications, while Table 9 shows the summary of Trade-offs in Neural Networks for threat detection.

2) BALANCING ENERGY EFFICIENCY AND CYBERSECURITY EFFECTIVENESS

Research into wireless sensor networks has demonstrated fundamental tensions between security effectiveness and energy conservation. Techateerawat and Jennings conducted pioneering work in this area, documenting how intrusion detection systems face inherent trade-offs between attack detection accuracy and energy consumption [188]. This finding has been further supported by Althunibat et al., who identified similar challenges in cognitive radio networks where cooperative spectrum sensing must balance energy efficiency against security measures designed to detect malicious users providing false information [189]. The implementation of AI-based threat detection systems introduces additional layers of complexity to these trade-offs. Empirical research by Tsikoudis et al. revealed that while low power designs can achieve up to 30% reduction in energy consumption, they typically introduce processing delays ranging from 100-500 milliseconds in threat detection response times [190]. This performance impact becomes particularly concerning in neural network applications, where Shumailov et al. discovered that adversaries can exploit ''sponge examples'' to deliberately maximize both energy consumption and latency, potentially degrading system performance by a factor of 30× [191]. These findings highlight the critical nature of balancing power efficiency with security responsiveness.

In the domain of IoT security, Taddeo and Ferrante proposed a systematic approach using the Analytic Hierarchy Process to optimize the balance between security requirements and resource consumption, effectively formulating it as a knapsack problem for algorithm selection [192]. Building on this work, Tappari introduced an innovative resource optimization technique utilizing the delay-term instruction application in cryptography coding, demonstrating a 15-20% reduction in resource utilization for lightweight security protocols [193]. However, this approach requires careful calibration to avoid compromising security effectiveness.

The smart grid domain provides particularly compelling evidence of these trade-offs. Research by Abdallah et al. documented how efforts to improve energy consumption efficiency through two-way communication systems simultaneously increased vulnerability to cyberattacks capable of manipulating energy demands and operational costs [194].

TABLE 9. Summary of trade-offs in neural network architectures for threat detection.

Architecture	Accuracy	Computational Overhead	Notes	References
Multi-Layer Perceptrons (MLPs)	∼99%	10^3 to 1.5×10^4 seconds	Baseline performance; overhead increases with model complexity	[181], [182]
Convolutional Neural Networks (CNNs)	Up to 100% (benchmark); ∼96.88% on SAMPLE dataset	Varies with model depth and connectivity	High accuracy on structured data; may degrade on complex, distributed datasets	[181], [182], [183]
Stacked Ensembles (RNN + CNN)	Up to 99.9% or higher	As low as 0.32 seconds	Combines strengths of multiple architectures; optimized implementations reduce overhead	[185]
Specialized Ensembles	High (not always explicitly stated)	Up to 16.6× reduction vs. software-only methods	Hardware-accelerated; tailored for specific tasks (e.g., hardware malware detection)	[187]
Hybrid Approaches (e.g., CNN-BiLSTM)	99.89% on NSL-KDD; 98.95% on UNSW-NB15	Generally higher than traditional methods; offloading can help	Achieves very high accuracy with increased computational complexity; optimization strategies are key	[185]

Their findings showed potential energy savings through smart grid optimization, but also identified increased attack surfaces that could lead to up to 30% higher operational costs if successfully exploited [194].

Edge computing presents a promising solution to mitigate latency concerns while maintaining security efficiency. By processing data closer to its source, edge computing significantly reduces latency, enhances data security, and optimizes bandwidth usage [195]. Muhoza et al. quantified that the implementation of AI algorithms directly on edge devices reduced energy consumption by 21% compared to cloud-based processing while maintaining detection accuracy above 97% [196]. The study specifically focused on real-time threat detection in IoT networks, testing more than 1,000 devices over a six-month period. Himeur et al. also researched this field by developing a hybrid edge-cloud architecture that achieved a reduction in overall energy consumption while maintaining sub-100ms response times for threat detection [197]. Their architecture demonstrated particular effectiveness in handling distributed denial-of-service (DDoS) attacks, with a 94% detection rate at one-third the energy cost of traditional cloud-based solutions [197].

In blockchain networks, Venkatesan and Rahayu conducted a comprehensive study of hybrid consensus algorithms, documenting specific improvements in both security and efficiency metrics [198]. Their consensus protocol enhanced with machine learning demonstrated a considerable improvement in trust metrics while reducing energy consumption compared to traditional proof-of-work systems [198]. The study included a detailed performance analysis on three different blockchain platforms and five different network configurations [198]. Similarly, in cloud computing environments, Brahmam et al. implemented bioinspired load balancing algorithms in a test bed of different virtual machines, achieving 30% improvement in migration efficiency while maintaining security integrity measures [199].

3) EMERGING TECHNOLOGICAL SOLUTIONS FOR SUSTAINABLE CYBERSECURITY

Balancing cybersecurity with sustainability requires strategic investments in energy-efficient security solutions while minimizing trade-offs in effectiveness. Pankajakshan and Maheshwari conducted quantitative analysis that revealed that organizations with proactive strategic approaches demonstrate significantly higher success rates in implementing cybersecurity [200]. Their structural equation modeling showed that comprehensive policies and employee participation are crucial mediating factors between organizational culture and successful adoption of technology [200]. This organizational perspective is complemented by the systematic framework of Ige et al. for protecting green infrastructure, which emphasizes the critical need to integrate technological, regulatory, and human factors in cybersecurity implementations [201]. Through their analysis of a decade of data (2014-2024), they established that interdisciplinary collaboration and enhanced regulatory frameworks are essential to maintain robust security while optimizing energy efficiency [201].

Energy companies are actively incorporating sustainability into their energy transition strategies by balancing environmental responsibility and energy security [202]. A balanced approach ensures that cybersecurity does not become a limiting factor in the adoption of sustainable energy. This perspective aligns with the findings of In et al., who empirically investigated the risk-return relationship of low-carbon

investments [203]. Their study revealed that companies with low-carbon product portfolios, particularly carbon-efficient companies, achieved significant shareholder returns [203]. This highlights that financial success and sustainability goals can in fact be mutually reinforcing. In fact, their analysis showed that an investment strategy focusing on carbon-efficient firms generated positive abnormal returns of 3.5-5.4% per year [203], further underscoring the financial benefits of aligning sustainability with business performance. Organizations must prioritize cybersecurity frameworks that align with these transitions while minimizing vulnerabilities in digital infrastructure [204].

VI. AI AND ML IN CYBERSECURITY

AI and ML have been integrated into various aspects of cybersecurity, to provide organizations with advanced tools to detect and respond to threats more effectively. These technologies can analyze vast amounts of data in real time, identify patterns, and make decisions that would be difficult or impossible for human analysts to achieve [205]. Traditional cybersecurity methods, which often rely on manual processes and rule-based systems, are not sufficient to handle the sophistication of modern cyber threats [206]. However, AI and ML have emerged as powerful tools to improve cybersecurity, offering capabilities that go beyond conventional methods by providing adaptive, scalable, and efficient solutions [207].

A. APPLICATIONS OF AI AND ML IN CYBERSECURITY

AI and ML have been at the forefront of cybersecurity innovation, allowing a variety of applications to strengthen defensive and offensive techniques in digital security.

- Threat Detection and Prevention: AI and ML are widely used in threat detection; they identify anomalies, detect malware, and recognize patterns associated with cyberattacks [9]. Research shows that artificial intelligence (AI) and machine learning (ML) are increasingly integral to cybersecurity, offering transformative capabilities in threat detection, response automation, and risk assessment [208], [209]. AI techniques such as machine learning and natural language processing enhance threat detection by automating the identification of anomalies and prioritizing threats, thus improving the efficiency and precision of cybersecurity measures [210]. ML is particularly crucial in the detection of network anomalies, significantly reducing detection and response times and helping to find proactive threats [54], [211].
- Response Automation: AI and ML improve cybersecurity by automating threat detection, analysis, and response [212]. They enable rapid identification of anomalies, such as unusual network traffic or user behavior, signaling potential cyber threats, and ML models detect malware by recognizing patterns from large data sets, including novel threats. And also indicate that applications AI and ML in cybersecurity enable the automation of repetitive tasks, improve threat detection, and improve response accuracy [213]. Furthermore, machine learning algorithms can analyze incident data to determine the most effective response strategies, reducing the time and resources needed to contain and mitigate threats [214].
- Risk Assessment: Risk assessment is another key application of AI and ML in cybersecurity, risk assessment provides dynamic, data-driven insights. They are integrated to address the complexity of cyber threats, automate the identification of vulnerabilities, predict potential attack vectors, and assess the probability that threats exploit weaknesses [215]. Through continuous monitoring and analysis, AI models can assess the risk posture of systems and networks in real time, adapting to evolving threats [216]. ML models are also used for risk assessment, where they evaluate the potential impact of various threats based on historical data and current conditions. These models can predict the likelihood of specific attacks and help organizations prioritize their security efforts [217]. By providing a data-driven approach to risk management, AI and ML enable organizations to make more informed decisions about their cybersecurity strategies [218].

B. EFFICIENCY AND EFFECTIVENESS IMPROVEMENTS

AI and ML have significantly improved cybersecurity operation efficiency and effectiveness through repetitive automation tasks, real-time threat detection, and predictive analytics that help prevent attacks before they occur. AI and ML improve efficiency by simplifying time-consuming operations such as data analysis, threat detection, and response to incidents [89]. Although AI and ML improve effectiveness by providing more accurate threat detection, reducing the number of false positives and ensuring that threats are identified and addressed immediately [9]. For example, advanced ML techniques, such as deep learning and reinforcement learning, improve the accuracy of threat detection and enable adaptive threat modeling, which allows the real-time identification of complex cyber threats [219]. Similarly, AI's role in automating repetitive tasks and speeding up threat detection and response has been recognized as a significant aspect in strengthening security postures against cyber attacks [220]. In addition, the integration of AI into Identity and Access Management (IAM) within cloud environments has shown significant improvements in user authentication and access control, with advanced hardware and software configurations leading to greater system precision [221].

VII. AI AND ML TECHNIQUES FOR REDUCING ENVIRONMENTAL IMPACT IN CYBERSECURITY

This section discusses the overview of AI and ML strategies to reduce environmental impact in cybersecurity based on

the articles carefully reviewed for this study, as shown in Figure 7:



FIGURE 7. Overview of AI and ML strategies for lowering environmental impact in cybersecurity.

A. ENERGY OPTIMIZATION IN CYBERSECURITY OPERATIONS

Energy optimization is a critical area in cybersecurity operations, as continuous monitoring, threat detection, and response can lead to significant energy use. However, AI and Machine Learning (ML) introduced techniques that can optimize energy usage in cybersecurity operations, making these processes not only more effective but also more energy efficient [222]. For example, a particular training run of a model such as GPT-3 can consume hundreds of megawatt hours of electricity, which represents the annual energy use of several hundred US households [223], which has a substantial carbon impact, particularly when using non-renewable energy. Thus, it is important to discuss some of the techniques for AI- and ML-driven energy optimization which include. Dynamic resource scaling: This technique dynamically adjusts the distribution of cybersecurity tasks between servers and data centers based on real-time analysis of network traffic and threat levels [17]. Artificial intelligence algorithms adjust the computational resources allocated to cybersecurity tasks based on real-time demand with various techniques as indicated in (Table 10). By continuously monitoring network traffic and threat levels, AI systems can increase resources during peak threat periods and decrease during low activity times, thus saving energy without compromising security [159]. Table 10 shows the dynamic resource scaling encompasses both the consolidation and workload management techniques. The table categorizes ''Dynamic resource scaling'' as a method that utilizes various techniques, including virtualization, consolidation, and workload management, to optimize resource utilization. Specifically, consolidation involves merging workloads onto fewer servers for better efficiency, while workload management ensures effective resource allocation based on demand transformations. Both techniques are part of the broader dynamic resource scaling approach, which aims to adapt computational resources in real time according to current network demand and threat levels, ultimately reducing energy consumption.

TABLE 10. Techniques for dynamic resource scaling.

ine **Method**	**Techniques**	**Usage**	**References**
ine Dynamic resource scaling	Virtualization	Allows several virtual servers to run on a single physical server, decreasing the overall hardware footprint and energy consumption.	[241]
ine	Consolidation	Merges workloads onto fewer servers for optimal resource utilization.	[242]
ine	Workload Management	Ensures effective resource allocation depending on demand transformations.	[243]
ine			

Implementation of energy-efficient algorithms: This involves building algorithms, using Low-Power Machine Learning models. Reference [26] explained that the models are specifically designed to operate with minimal energy requirements, making them ideal for deployment in cybersecurity systems where energy efficiency is a priority. By integrating ML models, cybersecurity operations can maintain high levels of security while significantly reducing their energy consumption [224]. AI-driven data analysis and filtering techniques are used to reduce the amount of data that needs to be processed and stored. By identifying and filtering irrelevant or redundant data, AI systems can focus computational resources on critical tasks, thus reducing the overall energy consumption [225]. Predictive maintenance and resource allocation AI and ML can predict when specific hardware or software components in a cybersecurity system will need maintenance or upgrades, preventing overuse or failure that can lead to energy inefficiencies. By proactively managing the lifecycle of these components, organizations can reduce the energy wasted on inefficient or failing systems [229].

B. RESOURCE MANAGEMENT AND OPTIMIZATION

Efficient resource allocation and management are essential in cybersecurity, where the balance of computational, human, and resource use can have a direct impact on both operational performance and environmental sustainability [230]. AI and ML provide significant tools for improving resources allocation and management in cybersecurity in order to ensure that resources are used effectively and enabling sustainable practices. One of the key applications of AI and ML in

resource management is to optimize resource allocation, using AI-driven systems to analyze the specific requirements of different cybersecurity tasks and allocate appropriate resources accordingly [231]. Another benefit is automated decision making, which can streamline cybersecurity processes by analyzing historical data and identifying patterns in resource usage [232]. According to Chataut et al. AI does not only predict future needs, but also recommend resource procurement and deployment strategies that minimize waste. Furthermore, the predictive analytics approach helps avoid the overuse of resources and reduces the environmental impact associated with their production, transportation, and disposal of excess resources [233]. Lastly, AI supports recycling of cybersecurity resources; for instance, AI algorithms can identify when certain software or hardware components can no longer serve their primary purpose but can still serve a secondary purpose, thus recommending their reuse or recycling [234]. This not only extends the lifecycle of components, but also reduces the rate of resource demand and further promotes sustainable practice in cybersecurity operations [235].

C. ENHANCED DECISION-MAKING AND POLICY DEVELOPMENT

The integration of AI and ML has become a known tool to improve decision-making processes and to establish cybersecurity policies that promote environmental sustainability. These technologies can analyze large datasets, predict future outcomes, and streamline challenging operations, making them valuable in the development and implementation of policies that reduce environmental impact while maintaining effective cybersecurity [236]. Enhanced decision-making and policy development green cybersecurity with the integration of AI and ML not only promote operation efficiency, but can also contribute to environmental sustainability while considering robust digital security [237]. AI and ML can improve policy development by providing data-driven insights into the environmental impact of different cybersecurity practices. The research by Nwokediegwu states that AI and ML algorithms can analyze a large amount of data from different sources, such as smart grids, environment monitoring systems, and cybersecurity incidents, to provide significant insights for policymakers and decision makers [238]. In addition, this technology can automate the analysis and enforcement of cybersecurity policies to ensure that organizations comply with environmental regulations. For example, AI-driven predictive analysis can continuously monitor energy consumption, detect changes in sustainable norms, and alert administrators to take appropriate action [239]. In practice, Perkumiene et al. developed an AI-driven decision tree model to classify environmental protective agency status based on emission parameters and achieved an impressive accuracy of 99.9% [240].

VIII. CHALLENGES AND LIMITATIONS

Having reviewed the integration of AI and ML into green cybersecurity for the promotion of sustainable development and protection of the environment, it is noticeable that it faces various challenges that must be addressed to ensure effective implementation. These challenges arise from the complexities of their application in cybersecurity systems, these challenges include:

1) High Computational Demand: The primary challenge of these technologies is the high computational demand for AI and ML algorithms. Training and implementing advanced AI models, particularly deep learning models, require huge computational resources, which can lead to increased energy consumption [241]. This is a significant challenge in green cybersecurity, where the goal is to reduce the overall environmental footprint. Researchers are currently exploring energy-efficient algorithms and hardware acceleration techniques to mitigate this issue. For instance, the development of Low-Power Machine Learning models aims to reduce the energy required for both training and inference processes. In addition, specialized hardware such as Tensor Processing Units (TPUs) and AI-optimized processors are being designed to enhance computational efficiency, potentially lowering the energy cost of AI-driven cybersecurity tasks [242].
2) Difficulty in integrating AI and ML with existing cybersecurity infrastructure: Many cybersecurity systems are built on older technologies that may not be easily compatible with modern AI and ML tools [220], [243]. This leads to integration difficulties, where the benefits of AI and ML are limited by the constraints of older systems. Researchers are also focusing on identifying middleware and integration frameworks that can bridge the gap between older systems and AI-driven tools. In addition, efforts are being made to standardize AI and ML applications in cybersecurity, which could facilitate smoother integration and broader adoption.
3) Data Availability and Quality: The issue of data availability and quality is also a significant challenge. To have effective AI and ML models, high quality data is required for their training, while such data may be difficult to obtain in the context of green cybersecurity [205]. This is particularly true for data related to energy consumption and resource utilization in cybersecurity operations, which may not be consistently recorded or made available for analysis. To overcome this challenge, researchers should utilize data augmentation techniques and the use of synthetic data to supplement real-world datasets. Additionally, collaborative efforts should be made to establish shared data sources that can help with the development and testing of AI models in green cybersecurity.

A. ETHICAL AND SOCIAL IMPLICATIONS

In addition to technical challenges, ethical and societal implications are another crucial challenge when implementing the AI model in cybersecurity. One of the most pressing ethical concerns is the bias and unfairness of AI and ML algorithms [244], [245]. AI systems are often trained on historical data, which might involve existing biases or inequalities. A notable concern is the potential for biased threat detection mechanisms to disproportionately flag activities from marginalized communities or regions with limited digital infrastructure, exacerbating existing disparities [246], [247]. This issue is further complicated in green cybersecurity, where energy-efficient AI models can prioritize computational efficiency over fairness, leading to discriminatory results in threat prioritization [248]. Addressing these concerns requires the adoption of frameworks such as the EU Ethics Guidelines for Trustworthy AI, which emphasize transparency and explainability [249]. Bias audits using adversarial testing and data collection in diverse datasets are essential to mitigate these risks [250], [251]. Furthermore, the development of fairness-aware algorithms that dynamically adjust decision thresholds based on contextual equity metrics ensures that benefits are equitably distributed [252], [253]. Lightweight AI models optimized for energy efficiency, such as pruned neural networks, must undergo rigorous fairness assessments to maintain ethical standards while achieving sustainability goals [254].

Another pressing issue is the impact of AI on the cybersecurity workforce. Automating routine tasks, including log analysis and threat monitoring, poses a significant risk of job loss [255]. Estimates suggest that by 2030, AI could replace approximately 20% of cybersecurity jobs, leading to a global talent shortage of more than 85 million workers [256], with the roles heavily reliant on manual data processing the most affected. According to a report by the McKinsey Global Institute, at least 14% of employees worldwide may need to transition to new careers due to the rapid pace of digitization, robotics, and AI-driven disruptions [257]. This shift threatens economic stability and the well-being of employees, as job insecurity can erode trust between workers and organizations, potentially reducing participation in environmentally conscious behaviors [258]. Strategies to address this challenge include large-scale reskilling initiatives where governments and corporations collaborate to train workers in AI ethics auditing, model transparency oversight, and green cybersecurity policy design [259]. Transparent communication from leadership on the role of AI in augmenting rather than replacing human decision making can alleviate job insecurity [260]. Implementing human-AI collaboration frameworks, where analysts oversee AI-generated insights and refine model logic, reinforces the complementary nature of AI and human expertise [261]. Additionally, social safety nets such as Universal Basic Income (UBI) trials, as proposed in South Korea's AI transition policies, could serve as models for balancing technological advancement with social equity [262].

Green cybersecurity also faces the challenge of balancing security with sustainability [263]. AI models designed for energy efficiency, such as quantized neural networks, can reduce energy consumption by up to 70% [264]. However, this efficiency gain may come at the cost of threat detection accuracy, particularly for new threats, which could introduce vulnerabilities into critical infrastructure [265]. Energy-optimized intrusion detection systems, for instance, may prioritize low power operations over real-time analysis, leading to delayed responses to zero-day exploits [190]. To navigate these trade-offs, adaptive AI models that adjust energy usage based on threat severity are crucial. The AI Cybersecurity Dimensions (AICD) framework proposes a tiered energy allocation strategy, where high-risk scenarios trigger full computational capacity while low-risk conditions conserve energy [266]. Furthermore, the intersection of privacy and energy efficiency remains a critical concern. Although lightweight encryption protocols for IoT devices are energy efficient, they may inadvertently expose sensitive data protection [267]. Adopting a privacy-by-design approach, as seen in GDPR-compliant AI systems, ensures that energy savings do not come at the expense of data [268].

Equitable access to cybersecurity tools remains a global challenge, and green cybersecurity initiatives often exacerbating the digital divide [269]. Regions with limited infrastructure face increased vulnerability, as energy-efficient systems tend to prioritize resource-rich areas [270]. For example, AI-driven grid security tools deployed in developed nations may neglect communities reliant on outdated hardware, making them more susceptible to cyberphysical attacks [270]. Policy interventions are necessary to bridge this gap. International funding for green cybersecurity tools in developing economies, modeled after the Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure), can help level the playing field [271]. Furthermore, the promotion of open source AI tools, such as Green AI, offers low-cost, energy-efficient AI models tailored for regions with intermittent power access, fostering inclusive adoption and strengthening global cybersecurity resilience [272].

Rapid advancement of A in cybersecurity has outpaced regulatory frameworks, creating accountability gaps [273]. The EU AI Act classifies cybersecurity AI as ''high risk,'' necessitating transparency reports and third-party audits [274]. However, global regulatory harmonization remains a challenge, with fragmented policies across the U.S. and Asia-Pacific regions. Establishing global ethical standards is imperative [275]. Adopting the OECD AI Principles for cybersecurity can provide a foundation for human oversight, accountability, and environmental sustainability [276]. Furthermore, implementing risk-based ethical

assessments such as the Ethics Impact Assessment (EIA) can ensure that AI systems are evaluated for bias, privacy violations, and carbon footprint before deployment [277]. As AI continues to shape the cybersecurity landscape, ethical governance and regulatory imperatives must evolve in tandem to safeguard both security and social equity [273].

In the context of cybersecurity, this could lead to biased decision-making, in which certain threats or vulnerabilities are prioritized over others based on biased information [278]. However, in efforts to reduce energy consumption, there is a risk that AI-driven decisions may affect certain user groups or regions, particularly those with less access to advanced technologies. To address these concerns, it is essential to develop transparent and explainable AI systems that can be audited for bias and unfairness. Furthermore, research should look at fairness-aware algorithms to reduce bias in AI decision making and ensure that all stakeholders benefit equally from advances in green cybersecurity. Another ethical concern is the balance between security and sustainability. Although AI and ML improve both cybersecurity and environmental sustainability, there may be times when this objective may contradict [163]. Minimizing energy consumption in cybersecurity operations can result in slower response times or reduced security coverage, affecting threat detection and response effectiveness. This raises the challenge of how to maintain the balance between the need for strong cybersecurity and the need for protection of the environment. Ethical guidelines and frameworks are required to navigate this balance, ensuring that AI system measures are aligned with both security and sustainability goals. Social effects are another challenge that should be equally considered, especially in terms of employment and skill development [279]. As AI and ML play more roles in cybersecurity, there is a risk of job displacement, particularly in jobs that require routine work. To avoid the negative impact on sociality, employers must participate in reskilling and upgrade programs that prepare employees for the evolving cybersecurity environment [280]. In addition, measures that promote equal access to AI technologies can help ensure that the benefits of green cybersecurity are widely distributed throughout society.

IX. RECOMMENDATIONS

Advancing green cybersecurity requires a strong emphasis on interdisciplinary collaboration, bringing together experts from diverse fields such as computer science, environmental science, engineering, ethics, and policy making. The complex challenges at the intersection of AI, ML, cybersecurity, and environmental sustainability cannot be effectively addressed alone. Instead, they demand the combined expertise of multiple disciplines [281].

A. POLICY RECOMMENDATIONS FOR GREEN CYBERSECURITY

- **Tax Incentives for Energy-Efficient Cybersecurity Practices**: Governments should introduce tax rebates, deductions, or subsidies for organizations adopting energy-efficient cybersecurity tools and infrastructure. This approach is related to China's successful subsidy programs for renewable energy which was studied by Li et al. From his research, it shows that tax reduction incentives can promote greener performance which in turn will improve green cybersecurity [282]. These incentives can stimulate R&D investment, boost technical efficiency, and facilitate technological progress in green companies [283].
- **International Standards Alignment (e.g., ISO 50001):** Adopt ISO 50001 as a global benchmark for green cybersecurity practices. Governments should require critical infrastructure providers (e.g., power grids, financial systems) to align their cybersecurity frameworks with ISO 50001's energy performance indicators (EnPI). The integration of ISO 50001 energy management systems can significantly improve energy performance indicators and efficiency in organizations [284]. This standardizes sustainability efforts while allowing flexibility in implementation, avoiding the pitfalls of rigid compliance-driven models.
- **Regulatory Mandates for Energy-Efficient Infrastructure:** Regulations could mandate compliance with ISO 50001 (energy management systems), requiring organizations to adopt its Plan-Do-Check-Act (PDCA) cycle to track and reduce energy use. This approach was also used by Lewry to carry out energy management within an organization [285]. This approach has been successfully applied in various industries, including deep-level gold mining, where it achieved an energy reduction of 18. 5% and significant cost savings [286].
- **Carbon Taxes on Energy-Intensive Cybersecurity Operations:** Government should Impose carbon taxes on firms that exceed predefined energy thresholds for cybersecurity activities. This policy is also related to a study by Singh et al. that suggested the use of IoT and blockchain technologies to track carbon footprints and impose dynamic taxes on industries that exceed carbon thresholds [287]. This approach reduces the cost of consumer trade and improves the overall well-being of organizations [288].

B. RECOMMENDATIONS AND FUTURE RESEARCH DIRECTION

AI and ML can affect cybersecurity practices, defining the future of green cybersecurity through several emerging trends, illustrated in Figure 8. One of the emerging trends is the development of AI-driven adaptive systems that are capable of modifying cybersecurity operations to reduce energy consumption in real time [289]. The systems use AI to monitor environmental variables, such as renewable energy sources, and adjust cybersecurity practices accordingly, thus optimizing both security and sustainability [289]. Another emerging trend is the analysis of quantum computing and

its possible application in green cybersecurity. Quantum computing can compute complex datasets at remarkable speeds, potentially lowering the energy required for AI and ML activities in cybersecurity [290]. Therefore, future research and development activities can focus on more energy-efficient AI and ML technologies to fully minimize the environmental impact of cybersecurity operations. In addition, more effective risk management techniques are needed to ensure long-term security and green cybersecurity practices.

- **Strengthen Interdisciplinary Collaboration** Future efforts in green cybersecurity should facilitate collaboration between computer scientists, environmental researchers, policymakers, and ethicists. Such collaboration can create a multidimensional framework for addressing both technological and environmental concerns effectively.
- **Adopt Energy-Efficient Cybersecurity Protocols** Organizations should prioritize the development and implementation of energy-efficient cybersecurity practices. By leveraging AI and ML algorithms designed for minimal energy consumption, companies can reduce their environmental impact without compromising their cybersecurity posture.
- **Invest in Training and Awareness** Cybersecurity professionals must receive training in sustainable practices and the applications of AI and ML. Enhancing awareness about the environmental implications of cybersecurity operations can motivate more ethical decision-making within organizations.
- **Implement Regulatory Frameworks** Governments and regulatory bodies should establish guidelines that promote green cybersecurity practices. Encouraging innovation through policies that support sustainable practices while ensuring robust cybersecurity can create a balanced approach to digital security.
- **Enhance Data Privacy and Security Standards** As organizations integrate AI and ML, it is crucial to prioritize data privacy and security. Developing standards that safeguard user data while utilizing these advanced technologies will not only build trust but also mitigate potential ethical dilemmas.
- **Exploration of Quantum Computing in Cybersecurity** Future research should investigate the potential applications of quantum computing within green cybersecurity frameworks. By harnessing the speed and efficiency of quantum computing, we may reduce the energy required for data processing significantly.
- **Develop AI-Driven Adaptive Systems** There is a pressing need to explore AI-driven adaptive systems that optimize cybersecurity operations based on real-time environmental conditions. To achieve this, researchers should focus on algorithms that can adjust cybersecurity measures according to energy availability and consumption metrics.
- **Comprehensive Life Cycle Assessment** Conducting extensive life cycle assessments of cybersecurity technologies can provide insights into their environmental impacts. Future studies should investigate not just the operational efficiencies of AI and ML but also the environmental costs associated with their entire lifecycle, from development to deployment and decommissioning.
- **Investigate the Socioeconomic Impact of Green Cybersecurity** It is essential to study the socioeconomic factors influencing the adoption of green cybersecurity practices. Understanding barriers such as cost implications, resource allocation, and industry readiness can inform better strategies for promoting sustainability in cybersecurity.
- **Quantum-Resistant Cryptography**: The quantum age is reshaping cybersecurity into a high-stakes race against time. To stay ahead, organizations must urgently adopt quantum-resistant cryptography–a cornerstone of green cybersecurity strategies. These solutions are critical for protecting blockchain networks and IoT ecosystems from future quantum attacks, which could exploit today's encryption methods. By acting now, industries can future-proof their systems before quantum computers evolve from theoretical risks to widespread threats.
- **Biologically Inspired Security Models**: Blockchain can secure peer-to-peer (P2P) energy trading in smart grids, but its potential in sustainable energy ecosystems remains underutilized [291]. For example, tokenizing carbon credits or renewable energy certificates on a blockchain can improve transparency and reduce fraud [292]. Researchers should focus on using smart contracts to automate energy transactions between IoT-enabled solar panels and grid stakeholders, reducing reliance on centralized (and often carbon-intensive) systems [293].
- **Quantum-Blockchain Symbiosis**: Researchers should work on designing blockchain frameworks that use quantum computing for threat detection (e.g., quantum-enhanced anomaly detection) while using energy-efficient consensus to offset quantum computing's own energy demands [294].
- **Policy and Regulatory Frameworks**: To promote sustainable practices, policymakers need to create global standards that encourage green cybersecurity. This could involve offering tax incentives for organizations that use energy-efficient hardware or imposing fines for high carbon footprints, promoting accountability.
- **Collaboration with Environmental Science**: Working with environmental scientists is key to improving green cybersecurity. Partnering with these experts can help organizations can create standard methods to measure the carbon emissions and energy efficiency of AI and ML systems. This ensures a clear and transparent assessment of their environmental impact.

- **Ethics and Governance in AI Usage** As AI and ML technologies continue to evolve, research must focus on developing ethical frameworks that govern their usage within cybersecurity contexts. This includes tackling issues related to bias, accountability, and transparency in AI-driven decisions.

C. ROLE OF INTERNATIONAL STANDARD IN PROMOTING GREEN CYBERSECURITY

ISO 50001 is a globally accepted Energy Management System that offers a framework for organizations to manage their energy consumption and reduce their carbon footprint [295], [296]. The standard was first published by the International Organization for Standardization (ISO) in 2011 and has been adopted by many organizations worldwide. The standard provides a systematic approach to energy management that can be integrated into existing management systems, making it easier for organizations to implement and maintain [296]. A study conducted by the Lawrence Berkeley National Laboratory found that organizations that implemented ISO 50001 achieved an average energy performance improvement of 10% within the first two years of implementation [297]. Here are the Roles of ISO 50001 in promoting Green Cybersecurity.

- **Providing A Structured Framework for Energy Management:** ISO 50001 establishes a systematic Plan-Do-Check-Act (PDCA) cycle to optimize energy use in cybersecurity operations [298]. This framework helps organizations set measurable energy reduction goals, implement efficiency measures (e.g., using renewable energy for data centers), audit outcomes, and refine processes [299]. The standard's iterative approach ensures continuous improvement, making it easier to align cybersecurity practices with sustainability targets.
- **Standardizing Energy Performance Metrics:** ISO 50001 mandates the use of Energy Performance Indicators (EnPIs), which provide quantifiable metrics such as energy consumption per terabyte of encrypted data or CO_2 emissions from security systems [300]. These metrics serve two critical functions: enabling organizations to benchmark their performance against industry peers and facilitating compliance with environmental regulations such as the EU Taxonomy. The standard's emphasis on continuous improvement through systematic data collection and analysis [301] has particular relevance for cybersecurity operations, where organizations can measure and optimize the energy efficiency of various security measures, from AI-driven threat detection to encryption processes. The ISO 50001 Impact Estimator Tool (IET 50001) provides organizations with the means to assess their implementation's global impact, including energy consumption, CO_2 emissions, and cost savings [302]. These metrics enable organizations to quantify the environmental impact of cybersecurity infrastructure and benchmark progress against other industries.
- **Driving Innovation in Energy-Efficient Technologies:** ISO 50001 encourages innovation in the research and development in green cybersecurity solutions, such as energy-efficient AI algorithms and sustainable hardware [303].
- **Building Stakeholder Trust Through Transparency:** ISO 50001 certification helps organizations build trust with their stakeholders by clearly showing their dedication to saving energy and protecting the environment. When a company gets this certification, it gains trust from different groups: investors feel more confident because they can see real data about energy savings, customers appreciate the company's environmental efforts, and employees can see that their workplace truly cares about sustainability [304]. Going beyond just following rules, the certification helps companies share their environmental progress openly and honestly, which is especially important in today's business world where ethics matter [299]. This openness helps companies build strong, lasting relationships with people and organizations who care about environmental responsibility [297].

D. ROLES OF HUMAN EXPERTISE IN GREEN CYBERSECURITY

Although AI and ML-driven tools optimize energy consumption and threat detection, their deployment, governance, and ethical application require rigorous human oversight. For example, AI models powering energy-efficient intrusion detection systems (IDS) require cybersecurity professionals to calibrate them to achieve optimal balance between accuracy and computational efficiency. Without skilled personnel to interpret outputs, mitigate false positives, or align systems with organizational sustainability objectives, even advanced algorithms risk inefficiency or misuse.

Human expertise is indispensable in ensuring that AI solutions adhere to green cybersecurity principles, which prioritize energy efficiency and environmental sustainability [305]. Cybersecurity professionals must implement robust defense strategies, including encryption protocols, access controls, and regular security audits, to protect critical infrastructure and ensure continuous energy supply and must also have specialized knowledge to assess the carbon footprint of AI tools, integrate renewable energy sources into cybersecurity infrastructure, and enforce ethical guidelines for resource allocation [130]. Table 11 shows the overview of Professional roles and responsibilities, in green Cybersecurity and Sustainability.

To bridge this skill gap, targeted training programs and certifications in energy-efficient cybersecurity practices are essential. These programs should equip professionals with competencies in sustainable technology integration, lifecycle analysis of AI systems, and governance frameworks for minimizing energy waste. Certifications such as the Professional Certificate in Green Identity Theft Prevention [306]

TABLE 11. Overview of professional roles, responsibilities, in green cybersecurity and sustainability.

Role	Responsibilities	Reference
Cybersecurity Expert	• **Design Secure Systems:** Develop and implement robust cybersecurity architectures and secure coding practices. • **Manage Risks:** Perform comprehensive risk assessments, threat analysis, and mitigation planning. • **Address Skills Shortage:** Initiate training, mentorship, and knowledge-sharing programs to bolster cybersecurity talent.	[308]
Environmental Scientist	• **Assess Ecological Impact:** Evaluate the environmental footprint of cybersecurity infrastructures, including e-waste and energy consumption. • **Explore Synergies:** Identify and promote mutually beneficial strategies between green IT and cybersecurity practices to enhance sustainability.	[309], [310]
Electrical Engineer	• **Assess Vulnerabilities:** Identify and evaluate vulnerabilities within electrical grids. • **Analyze Potential Failures:** Investigate and simulate potential failures and disruptions. • **Implement Risk Mitigation Strategies:** Develop and execute strategies to enhance grid resilience.	[311]
Policy Analyst	• **Develop Robust Defense Strategies:** Formulate policies for information classification, access control, and data encryption. • **Promote Security Awareness:** Organize and support security awareness training initiatives. • **Plan Incident Response:** Develop and refine incident response procedures to mitigate cyber threats effectively.	[312]

and the Postgraduate Certificate in Carbon Offset Projects for Cybersecurity–both offered by the London School of Business and Administration [307] –could standardize expertise, ensuring that practitioners align technological innovation with global climate goals.

X. CONCLUSION

As we stand at the crossroads of digital innovation and environmental sustainability, this study underscores the critical importance of integrating green cybersecurity practices within our technological frameworks. The research findings reveal that while Artificial Intelligence (AI) and Machine Learning (ML) enhance cybersecurity operations, they also provide avenues to reduce carbon footprints and overall energy consumption. These technologies are not just tools for defense against cyber threats; they are also vital components in our collective effort to achieve sustainability in the digital age.

The study demonstrates that the intersection of cybersecurity and sustainability is burgeoning with potential, driven by the adoption of AI and ML. For example, advanced adaptive systems can dynamically adjust cybersecurity protocols in real time based on renewable energy availability, showcasing a model where security measures align with environmental considerations. This dual focus is paramount as organizations grapple with the growing need to fortify their cyber defenses while simultaneously addressing climate change.

However, despite the promising prospects, the journey towards effective implementation of green cybersecurity is fraught with challenges. The high computational demands of AI and ML algorithms pose significant barriers, particularly concerning energy use and the carbon emissions associated with running these systems. Moreover, the lack of interdisciplinary collaboration hinders the development of holistic solutions that encompass the necessary expertise from fields such as computer science, environmental science, policy-making, and ethics.

To move forward, this research highlights the need for a concerted effort to address these challenges through focused

research and innovation. Future studies should aim to refine AI and ML technologies for improved energy efficiency, thereby minimizing their environmental impacts. In particular, examining the potential of quantum computing within the realm of green cybersecurity could lead to breakthroughs in processing capabilities while reducing energy consumption.

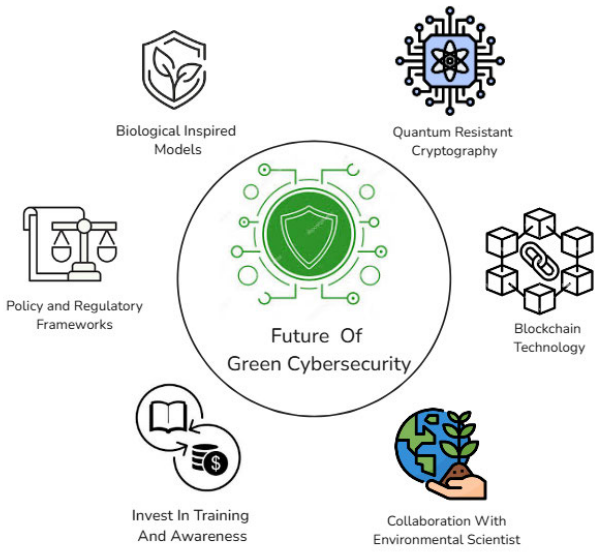


FIGURE 8. Future directions of green cybersecurity.

AUTHOR CONTRIBUTIONS

The authors declare that they have equally contributed to the article. All authors read and approved the final manuscript.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

ABBREVIATIONS

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
ML	Machine Learning
LLMs	Large Language Models
SDGs	Sustainable Development Goals
CMM	Cybersecurity Maturity Model
LCA	Lifecycle Assessment
DDoS	Distributed Denial-of-Service
CPS	Cyber-Physical Systems
NLP	Natural Language Processing
MWh	Megawatt-hours
DVFS	Dynamic Voltage and Frequency Scaling
IoT	Internet of Things
UEBA	User and Entity Behavior Analytics
SIEM	Security Information and Event Management
CVE	Common Vulnerabilities and Exposures
FFT	Fast Fourier Transform
DNNs	Deep Neural Networks
QNNs	Quantized Neural Networks
AutoML	Automated Machine Learning
YOLOv5	You Only Look Once version 5
Faster R-CNN	Faster Region-based Convolutional Neural Network
MNIST	Modified National Institute of Standards and Technology database
CIFAR-10	Canadian Institute for Advanced Research - 10 classes
SVHN	Street View House Numbers
ImageNet	A large visual database designed for use in visual object recognition software research
TPUs	Tensor Processing Units
IAM	Identity and Access Management
GPT-3	Generative Pre-trained Transformer 3
BERT	Bidirectional Encoder Representations from Transformers
APQ	Architecture, Pruning, and Quantization
ACTD	Accelerate CNNs from Three Dimensions
IntraQ	Intra-class Quantization

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